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**SMART INVENTORY SYSTEM:
SALES ANALYSIS USING
SARIMAX FOR A CHEMICAL
MANUFACTURING COMPANY**

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APPROVAL SHEET

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ABSTRACT

Smart inventory management systems are mainly used by many businesses to track real-time stock. When tackling daily sales, time series analysis algorithms are favored over Neural Networks. The proponent Emdee Chemical Marketing Corporation is currently facing problems of not having a digital inventory system to track their data. Furthermore, SARIMAX, a time-series forecasting algorithm, offers a promising approach for forecasting sudden bulk sales of a product. This study employs Rapid Application Development (RAD) software methodology to iteratively design and implement the Smart Inventory Management System (SIMS) features. The web-based application is built using Python, TypeScript, and SQL, and is deployed in the cloud.

It features two core functionalities: inventory management and sales forecasting. The forecasting module utilizes internal sales invoice data for predictive analytics, enabling the business to anticipate fluctuating demand and optimize inventory levels. The research utilized questionnaires with Likert-scale questions and open-ended questions to evaluate the web-based application to its target beneficiary. SARIMAX was relatively accurate in predicting the sudden increase of sales since it can predict the trendline of it going up, but not to the extent of the original sales. When scaled down the lowest or most accurate split is the 80/20 split with scores of MAE, RMSE, and MAAPE of 0.27, 0.33, and 0.2078. Overall, Emdee Chem believes with the integration of the web-based application into their day-to-day operations they would have increased productivity and inventory accuracy.



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CHAPTER I

INTRODUCTION

1.1 PROJECT CONTEXT

Inventory management systems are necessary for businesses to keep track of their supplies, maintain their inventory at optimal levels, and check availability at all supply chain points. Mishandling inventory would waste resources, delay deliveries, and decrease productivity (OptiProERP, 2023). Spreadsheets, manual order placements, and hand-counted stock levels are all part of the inventory management system. With all of these combined the system can automate end-to-end, business management, forecasting, and accounting leading to the simplification of the processes of ordering and storing (IBM, 2023).

Smart inventory management systems are used by businesses for real-time stock tracking, thus preventing under or overstocking of their inventory (Webb, 2024). The implementation of advanced intelligent features into inventory management such as machine learning techniques, advanced algorithms, and automated features can potentially improve a business by streamlining operations, maximizing productivity, and meeting customer demands.

Based on a study by Gordon (2021), sales forecasting is important for all businesses because it allows the company to make more informed business decisions such as



inventory management, and budgeting. Sales forecasting helps with resource allocation using an estimation of a company's sales over a certain time. Applying this to a Smart Inventory System would improve the decision-making and resource allocation of a company.

Time series analysis is a statistical method used in data collected sequentially over time. These are used to understand patterns, trends, and behaviors to make forecasts about future events. Incorporating machine learning in time series analysis leads us to the many different machine learning models. These models include neural networks (Lin et al., 2019), tree-based methods such as decision trees and random forests (Collado & Tumibay, 2023), and Autoregressive Models (Ichon & Gente, 2023). These models include Autoregressive Moving Average (ARMA) models and their extensions, such as Seasonal Autoregressive Integrated Moving Average Exogenous (SARIMAX).

SARIMAX is an extension of ARMA that incorporates seasonality, integration, and exogenous variables. Seasonality is the inconsistent pattern that appears periodically. These may come in the form of holidays, seasons of the year, and even events such as market crashes and pandemics (Mehandzhiyski, 2023).

Emdee Chemical Marketing Corporation, or Emdee Chem for short, is a chemical marketing company that was established in Quezon City in 1997 and through its dedication to serving the needs of different industries has become a vital part of their operations by



supplying the production chemicals needed for their day-to-day operation. The mission of the company is to continue being the trusted choice for its quality products and services, helping other industries thrive in today's ever-changing world. Emdee Chem boasts a zero-delay on its deliveries and is recognized by multinational companies. Emdee Chem offers a variety of products ranging from automotive needs and chemicals, to housekeeping supplies, and even cosmetic goods such as creams, lotions, and more.

1.2 PURPOSE AND DESCRIPTION

The project's goal is to help the chosen business with its day-to-day operations and archive important data through the design and implementation of a smart inventory system. The benefits of a smart inventory system are inventory optimization, detection of supply chain disruptions, real-time inventory visibility, and higher customer satisfaction (Legchekov, 2022).

The project is a web-based application accessible online so that the chosen proponent can access the system at any place and at any time. The system included a database written in Postgres Standard Query Language (PostgreSQL), with the implementation of a SARIMAX model for sales forecasting, and automatic invoice generation based on user input. Lastly, the system has encryption security for the safety of the data of the Chemical Manufacturing Company.

1.3 STATEMENT OF THE PROBLEM



Emdee Chem makes use of simple spreadsheets, handwritten inventory procedures using logbooks, and manual processes to manage their inventory. This system has problems mainly with archiving as multiple departments of the company struggle to find the right record immediately.

The business admins of the company have stated that they are in need of an inventory system that keeps track of their incoming and outgoing products. Additionally, the admins have stated that archiving of previous records is a problem, especially in their inventory, there have been cases where raw materials inside the warehouse expire thus resulting in loss of profits (see Appendix H for the transcription of the interview).

Specifically, the research problems for the study are:

1. How can the researchers solve the slow inventory turnover experienced by the beneficiaries' warehouses?
2. Does the proposed system improve the inventory management experience in the conduct of business operations?
3. How effectively can SARIMAX capture and forecast the unique patterns of bulk and intermittent sales trends?
4. Once deployed, how would the sales analytics affect the admin's decision-making in manufacturing more products?



5. What would the changes in the performance of the company be if the system was deployed as a web application?

1.4 HYPOTHESIS

The researchers hypothesize that the implemented inventory management system with SARIMAX will improve Emdee Chem's day-to-day operations, by ensuring that the deliveries are always on time.

1.5 RESEARCH OBJECTIVES

1.5.1 General Objective

To develop a smart inventory management system that uses SARIMAX for a chemical manufacturing company.

1.5.2. Specific Objectives

- a. To interview the company to assess its needs through emails and Viber chats and onsite interviews at their main office.
- b. To list the inventory problems the company faces during the interview.
- c. To gather invoices data before developing the system.



- d. To design a database schema from the data gathered for the inventory system.
- e. To extract Excel sales data into a proper SQL database.
- f. To use the invoice sales data for training the model.
- g. To modify SARIMAX algorithms to be able to predict bulk and intermittent sales trends by using auto_arima.
- h. To implement sales analytics and sales growth reporting features using SARIMAX as its foundation.
- i. To integrate smart features such as autofill and stock tracking to automate day-to-day operations
- j. To create an intuitive and simple interface for users and administrators.
- k. To deploy the system using LightNode.
- l. To compare and measure efficiency before and after the new system was implemented using survey evaluation.
- m. To evaluate the system's performance and find out if there are future improvements that could be implemented for future research.

1.6 SIGNIFICANCE OF THE STUDY

The significance of this study lies in developing a system that the admins of the company may use to make data-driven decisions to reduce costs and make the process of storing data, products, and shipping them out much more efficient.



Furthermore, the study could be of importance to the following:

Emdee Chemical Marketing Corporation. The workflow process will become more streamlined thus making all the processes of the business from inventory counting to delivery and receiving more efficient. This study will also improve upon the current system of using a logbook to track their inventory and sales. Lastly, the proposed system will allow seamless management of inventory on a cloud database.

Clients of Emdee Chem. The clients of the proponents will receive better service due to the improved communication and movement of items all thanks to the implementation of the inventory system.

Future Researchers. Future researchers may use this study to improve upon the existing system or as a reference to their own work. This study can also serve as literature for smart inventory, machine learning, and sales forecasting. Furthermore, the study will be the foundation of using SARIMAX to predict intermittent and bulk sales. Since the data deals with sales ranging from thousands to millions, this study will be able to be the steppingstone for improving SARIMAX for predicting sales values of this magnitude.

1.7 SCOPE AND LIMITATIONS



The study focuses on developing and implementing a smart inventory management system for Emdee Chemical Marketing Company, the chosen Company located at 16, Dividend Street, GSIS Village, Project 8, Quezon City, 1106 Metro Manila. The system makes use of a SARIMAX model trained for forecasting future sales on the invoice data of the company spanning from 2020 - 2024. The system will be deployed as a web application that allows the employees of the company to access the dashboard and view the inventory of the products and customers, while the admins have access to the invoices, accounts, sales forecasting module, and access to creating, editing, and deleting entries from the database. The system is designed to improve the day-to-day operations of the company and improve customer experience by making sure that their deliveries are on time.

However, the study has several limitations. The forecasting model is limited to only using SARIMAX, which is not designed for intermittent sales forecasts as it requires a steady stream of data over a long period of time for accurate results. This system is limited to a web application and its requirement of a stable internet connection for access to the database. The system is also limited to the modules of product, user, customer, and invoice management for inventory and sales forecasting for the analysis. Additionally, the data for the system is limited to the marketable goods of the company and does not include raw materials. The data is also limited to the industrial side of the company and does include their cosmetic goods and other products. Lastly, the evaluation of the system is limited to the admins of the company that are experienced with using the traditional system and who



will be using the smart inventory system in their day-to-day operations. By acknowledging these limitations, the study aims to provide a transparent evaluation of the smart inventory system, highlighting areas for future research and potential improvements.



CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter is dedicated to discussing literature that encapsulates the different modules of the study and the system. It has two main parts, Local Literature and Foreign Literature.

2.1 FOREIGN LITERATURE

INDUSTRY 4.0

In recent years, the implementation of Artificial Intelligence (AI) and Machine Learning (ML) algorithms with information and communication technologies such as the Internet of Things (IoT) in businesses has led to smart technologies (Chung, 2021). Smart technologies have led to the period known as the Fourth Industrial Revolution widely known as Industry 4.0. This period is marked by making the process of producing goods smarter and more sustainable.

As reported by Yuen & Wu (2022) the benefits of Industry 4.0 are “real-time integrated process optimization; reduced delivery time by advanced forecasting approaches; new transport concepts to manage the last mile delivery efficiency; leverage the human planner for the disruptive events” (p. 1).



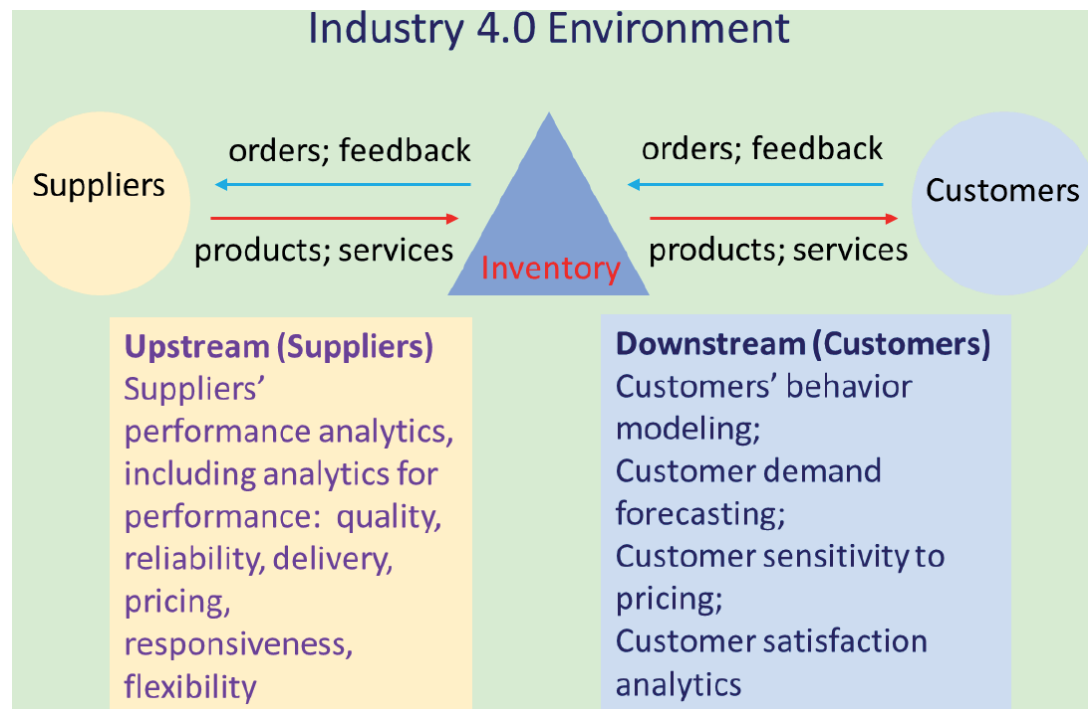
Businesses under Industry 4.0 have incorporated ideas of digitization and automation to stay competitive. To stay ahead of the competition these businesses must establish an information interchange with the use of data networks, automatic identification, and information communication technologies (Balfaqih, 2023). The benefits of Industry 4.0 to business are increased productivity and revenue, optimized manufacturing processes, better customer service, and better inventory management (Yuan, 2020).

Yuan (2020) stated that inventory management is greatly impacted by Industry 4.0. The impact can be summarized in four aspects. These aspects are: (1) Digitalized and automated purchase process and inventory forecasts for the customer needs through data analytics and business intelligence. (2) Improved inventory classification as the products can be segregated into different classes for ease of access and improved security. (3) Real-time sharing of location as the supplier and the business can track the location of the product and share the information in real time. (4) Better system reviews as the continuous review done by companies to monitor and review the inventory over time is improved by Industry 4.0 by sharing all available information in real-time making it convenient for companies to conduct continuous reviews.

In the same article, Yuan (2020) proposed an integrative research and development framework for the modeling of inventory systems and optimization as shown in Figure 2.1.1.

**Figure 2.1.1.**

Integrative R&D framework for inventory systems modeling and optimization





Retail companies must manage their resources and make strategic decisions to succeed. Since the market today has become more competitive, the pressure to increase profit margins has increased the need for sales forecasting. Especially sales forecasting methods using machine learning methods. Martins and Galeale (2023) evaluated three main methods in sales forecasting: Time Series, Artificial Neural Networks (ANN), and Machine Learning Algorithms. They concluded that machine learning has increased the accuracy of the forecasts, especially when the models contain exogenous and endogenous variables. Krishna et al. (2018) compared GradientBoost and AdaBoost algorithms for sales forecasting of retail stores and concluded that GradientBoost is the better predictor with its lower Root Mean Squared Error (RMSE) and higher predicted R-squared value.

There are two main approaches when creating sales forecasts and that would be a manual and data-driven approach. The data-driven approach is the approach where historical or time-series data. According to Xu (2023, p. 1688–1689), most companies create their time series data and algorithm to create a forecast for future sales by utilizing the time series data and consumer data. Xu (2023, p. 1688–1689), recommends that utilizing multiple models would reduce the deficiencies of a single forecasting model or algorithm.

Combining multiple models would improve the accuracy of the data. However, this is not always the case because there are models that only work on linear time series data such as ARIMA. In a study by Abbasimehr et al. (2020), they utilized an LSTM network with



multiple layers and compared it to other algorithms such as ARIMA and ETS. Their study showed that the LSTM network was the superior method in the demand time series analysis. They suggest using attention-based neural networks when creating a forecast from a time series (Abbasimehr et al., 2020).

The hardest type of demand to predict and create a model for is intermittent demand. Due to the random demand intervals throughout obsolescence, it is hard to create an accurate forecast model for slow-moving products with random demand intervals (Babai et al., 2020). Having slow-moving products increases costs due to holding costs and other miscellaneous costs like disposal. Based on a study by Babai et al. (2020), it is recommended to implement Croston's method followed by their method which is a modified SBA method when dealing with inventory with intermittent demand. The modified SBA method they used is a method that updates its demand intervals, size, and estimator only during periods with demand, and in periods without demand the updates would be similar to a probability occurrence used by the TSB method (Babai et al., 2020).

MAE, RMSE, and MAAPE are used to calculate the accuracy of the prediction because MAE and RMSE are scale dependent metrics typically used in ML accuracy evaluation (Jierula et al., 2021). MAAPE is used over MAPE due to it being a scale independent metric, while reducing the infinite and undefined values present. MAAPE is also able to implement the pros that are typically present in the MAPE (Algabroun & Håkansson, 2024).



2.2. LOCAL LITERATURE

Apolino & Norona (2021) assessed the issues that result in poor inventory accuracy and found that manual counting and the use of manual count sheets are inaccurate because it is prone to human error. To address these issues, they recommended automating the process using barcoding and implementation of Industry 4.0 to the warehouse operations and supply chain processes.

To remedy the issue, Aritonang and Sihombing (2019) compared two forecasting methods in determining the sales of the Rice Milling Unit (RMU). The two forecasting methods were Linear Regression and ANN with a Backpropagation algorithm. The research concluded that the ANN is more optimal with a Mean Squared Error (MSE) of 0.00099713 compared to the Linear Regression's MSE of 0.214.

Collado & Tumibay (2023) used tree-based machine learning models to forecast onion armyworms in Nueva Ecija. Random Forest and Decision Trees were used on the dataset provided by the local government. Additionally, binary classifiers and feature importance were used alongside the models to yield satisfactory results. Decision Tree yielded an accuracy of 88% and Random Forest had an accuracy of 89%.

Torculas et al. (2023) used XGBoost, Random Forest, Linear Regression, and Support Vector Regression for the forecasting of energy consumption in the Philippines. These machine-learning models were tested on monthly and quarterly assessments of the



Wholesale Electricity Spot Market (WESM). The study found that XGBoost had the most accurate predictions during the pre-pandemic period and Random Forest had the lowest RMSE for the pandemic period. The research recommended the use of Autoregressive models such as ARIMA for future uses.

ARIMA MODELS

Urrutia et al. (2019) used ARIMA and Bayesian Artificial Neural Network (BANN) to forecast the imports and exports of the Philippines and to use the forecasting accuracy measures such as MSE, NMSE, RMSE, and MAPE to find out which of the two is the most fitted model. The researchers concluded that the BANN is the most fitted model for the imports and exports of the Philippines. However, Delima (2019) used ARIMA to forecast the electric consumption of the Philippines. Using three different sectors and these are residential, commercial, and industrial use in the years 2018-2022. Delima (2019) determined that ARIMA is the best fit for the prediction of the Philippines' electric consumption.

Ichon and Gente (2023) used SARIMA to analyze the temperature in Davao Oriental from 2010 to 2022. SARIMA was chosen for its feature that incorporates seasonal and non-seasonal factors, and the leading model found was the SARIMA (0,1,3) (2,0,0) because it predicted the values of temperature with a 95% confidence interval.



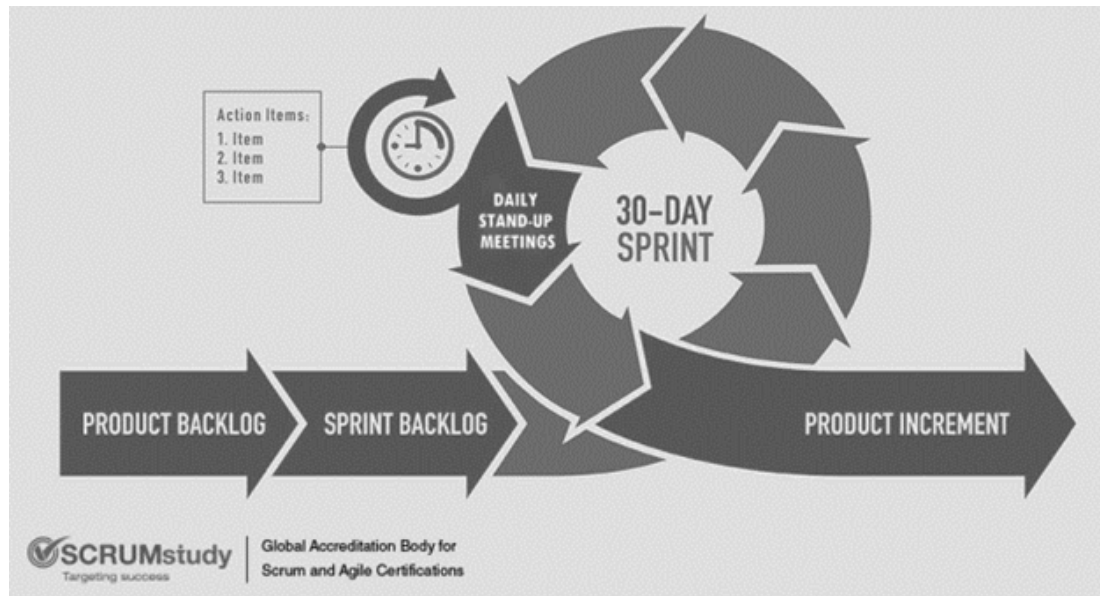
SARIMA can also be used to forecast the inflation rate of the Philippines, using 135 observations from January 2022 to March 2023 the model predicted that the inflation will gradually decline at the start of 2024 (Corpin et al., 2023).

LOCAL METHODOLOGY

Ng et al. (2019) used Applied Research methodology and Agile Driven Design to implement their inventory management website to replace GMI Company's manual methods of inventory, sales, and logistics system. Their study used an agile scrum methodology as seen in Figure 2.2.1. As shown in Figure 2.2.1, their development time will only take 30 days and will be divided into sprints.

Figure 2.2.1.

Agile Scrum Methodology.



Camaclang et al. (2022) used a descriptive research design for their research on web-based appointment systems. Their study utilized a questionnaire based on the ISO 25010:2023 standards, which asked their respondents about the system's security, operability, and functional requirements.



CHAPTER III

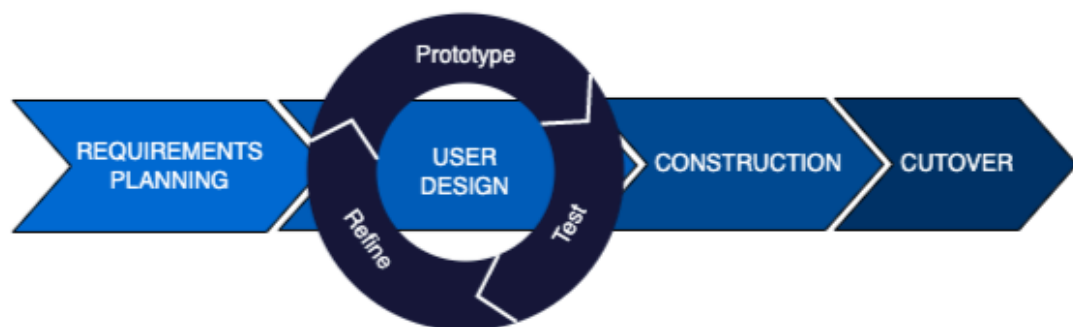
RESEARCH METHODOLOGY

This chapter discusses the software development methodology, conceptual and operational framework, and its algorithms, which are used to plan, analyze specifications, design, code, test, and maintain the Smart Inventory System. The aim is to determine a methodology to be chosen based on the technical and scope considerations needed to develop the system.

3.1 SOFTWARE DEVELOPMENT METHODOLOGY

Figure 3.1.1.

Rapid Application Development of Smart Inventory System.





The Rapid Application Development (RAD) shown in Figure 3.1.1, is an agile software development methodology that provides rapid development of a working software system. Saeed et al. (2019), agile methodologies like scrum, are less focused on long-term planning and more on development which is essential for the company's requests. This development methodology's advantage is, first, suited for small to large businesses, and second, being an iteration-based model, meaning development and testing can be done simultaneously for each iteration of each feature, as well as feedback from the client can be received immediately after each development cycle.

The first phase of RAD would be called Requirements Planning which focuses on analyzing and quickly designing the software specification. The next phase is User Design, which involves prototyping, testing, and refining the model of the system. Third, is Construction where the system, developed from initial models is integrated and fleshed out for final testing and feedback. Lastly, the Cutover phase, sometimes called Implementation, is where the application is deployed as the client deems it fit for deployment as well as the changeover for the new system.

3.1.1 REQUIREMENTS PLANNING

The Requirements Planning phase allows the system owner and the researchers to closely work together, research related literature for context and background, and gather sales and inventory data from the company's recorded sales history. This step allows for



revisions to be made to the plan. Throughout the development cycle, this step is the backbone of the project. The initial ML models (SARIMAX and XGBoost) for sales forecasting it is developed using Jupyter Notebook or Google Colab. As for building the interactive web application, it is developed using Express.js web framework, and PostgreSQL as its Database Management System (DBMS).

3.1.2 User Design

In the early stages of the User Design phase, user requirements are gathered through inquiring about the needs of the client. Once the requirements are known, the researchers immediately start developing the web application, implement its basic features, interface, and functionalities, and create an initial version of a working prototype. The researchers will then ask the clients for early feedback on what needs improving and progressively iterate, refine, and test current features until the client is satisfied with their requests for the system. Additionally, during this stage of development, the researchers identified the relationship between the company's product and customer data by implementing an Entity-Relationship Diagram (ERD) by utilizing a DBMS, as this is crucial to develop the data models for the system's database.

3.1.3 Construction

In the Construction phase, the different parts and features of the initial inventory system and its database are integrated by developing the API endpoints for the best-suited



ML model with the inventory system, allowing for real-time sales forecasting based on current inventory levels. Final tests are done on a local environment (localhost) for the different features to ensure and validate their functionalities.

3.1.4 Cutover

In the Implementation stage, with the features added and tested, the system is deployed in a production environment (cloud server) for the changeover of the previous manual system of sales. Also, the researchers guide and train users on how to adapt to the new system, providing documentation for its usage, and closely monitoring the system's performance.

3.2 CONCEPTUAL FRAMEWORK

Figure 3.2.1.

Conceptual Framework of the Study

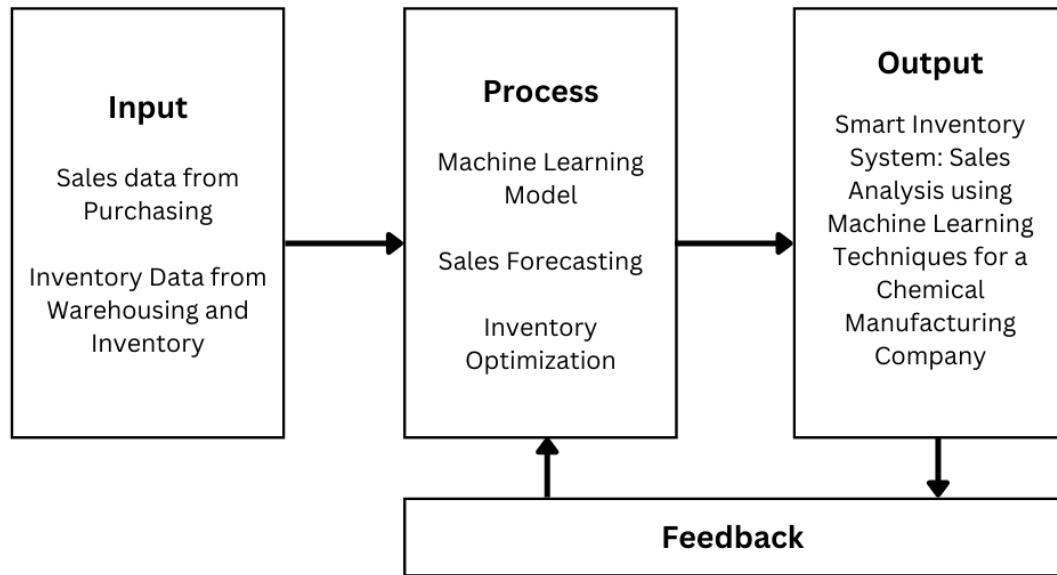


Figure 3.2.1. showcases the conceptual framework of the study. The IPO diagram describes the developmental phases of the study. The start of the development cycle requires data from the company. The company's sales data are used for the development and training of the SARIMAX model. The inventory data will be used for the inventory optimization and smart features for the system.

The data is used for the SARIMAX model, which forecasts future sales of the company and optimizes the inventory. After determining the most suitable forecasting technique, then the development of the Smart Inventory System will commence.



After the deployment of the system, the feedback of the company is used to further improve the system. Additionally, information about the accuracy of the model is also used as feedback.

3.3 ALGORITHM AND ITS RULES

Autoregressive Integrated Moving Average (ARIMA)

ARIMA(p,d,q) is a statistical analysis model that predicts future trends by using historical data, time series data, and similar datasets (Urrutia et al., 2019). Delima (2019) stated that the p,d,q denotes the autoregressive order ARIMA model can be represented in an equation as.

$$\phi(B)(w_t - \mu) = \theta(B)a_t$$

where,

t = time index

$\phi(B)$ = autoregressive parameter

B = backshift operator

$\theta(B)$ = moving average

w_t = d value in the ARIMA(p,d,q) model

a_t = white noise



Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX)

SARIMAX is an extension of ARMA that implements seasonality, integration, and exogenous variables. Khachatryan (2022) specified for time series the model of SARIMAX is:

$$\text{SARIMAX}(p,d,q)*(P,D,Q)_s$$

where,

p = autoregressive order

d = difference order

q = moving average order

P = Seasonal autoregressive order

D = Seasonal difference order, $D = 1$ would calculate a first order seasonal difference

Q = Seasonal moving average order, $Q=1$ would calculate a first order errors in the model

s = Seasonal period, $s=12$ due to a year having 12 months

3.3.1 FORMULAS AND MATHEMATICAL REPRESENTATIONS

Mean Absolute Percentage Error (MAPE)



According to Nik (2022), MAPE is used to measure the accuracy of a model. Specifically, MAPE defines the error of a given model by showing its loss function. The formula for MAPE is

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right|$$

where,

Σ indicates to add the resulting values,

N is the sample size,

A is the actual value,

F is the predicted value,

In MAPE the lower the value the better the model is at predicting values, see Table 3.3.1 for the interpretation of scores (Syaharuddin et al., 2023).

MAPE Score	Interpretation of Score
> 50%	Poor
20% - 50%	Relatively Good



10% - 20%

Good

<10%

Great

Table 3.3.1 *Interpretation of MAPE scores*

Mean Arctangent Absolute Percentage Error (MAAPE)

An alternative to MAPE, the Mean Arctangent Absolute Percentage Error (MAAPE) can evaluate the accuracy of a forecasting model as it serves the same function as MAPE but avoids the issue of dividing by zero when the actual value is zero. This metric measures the average magnitude and direction of the errors between the forecasted and actual values (Thieu, 2021).

Mohamad (2019) defined the formula of MAAPE as:

$$MAAPE = \frac{1}{N} \sum_{t=1}^N \arctan\left(\left|\frac{y_t - f_t}{y_t}\right|\right)$$

where,

y_t is the actual outcome at period t



$\mathbf{f}_t$ is the forecast value at period t

Root Mean Square Error (RMSE)

Root Mean Square Error is commonly used for evaluating the quality of predictions. It expresses how far the predictions stray away from the measured true values through Euclidean distance (Abbasimehr, et al., 2020) and it is expressed as.

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}$$



where,

n is the number of observations,

y_t is the observed value of the i -th observation,

$\hat{y}_t$ is the predicted value for the i -th observation,

Σ denotes the sum across all observations,

Mean Absolute Error (MAE)

Mean Absolute Error is the context of forecasting is the statistical measure that averages out the distance between the actual value and the predicted value. This metric treats all error equally, provides a clear and direct interpretation, and is less sensitive to outliers as stated by Singh (2023). In that same article, Singh (2023) describes the mathematical formula of MAE as.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Where,

n is the number of observations

y_i is the actual value of the i^{th} observation

$\hat{y}_i$ is the predicted value of the i^{th} observation





3.4 OPERATIONAL FRAMEWORK

Figure 3.4.1.

SARIMAX Project Operation.

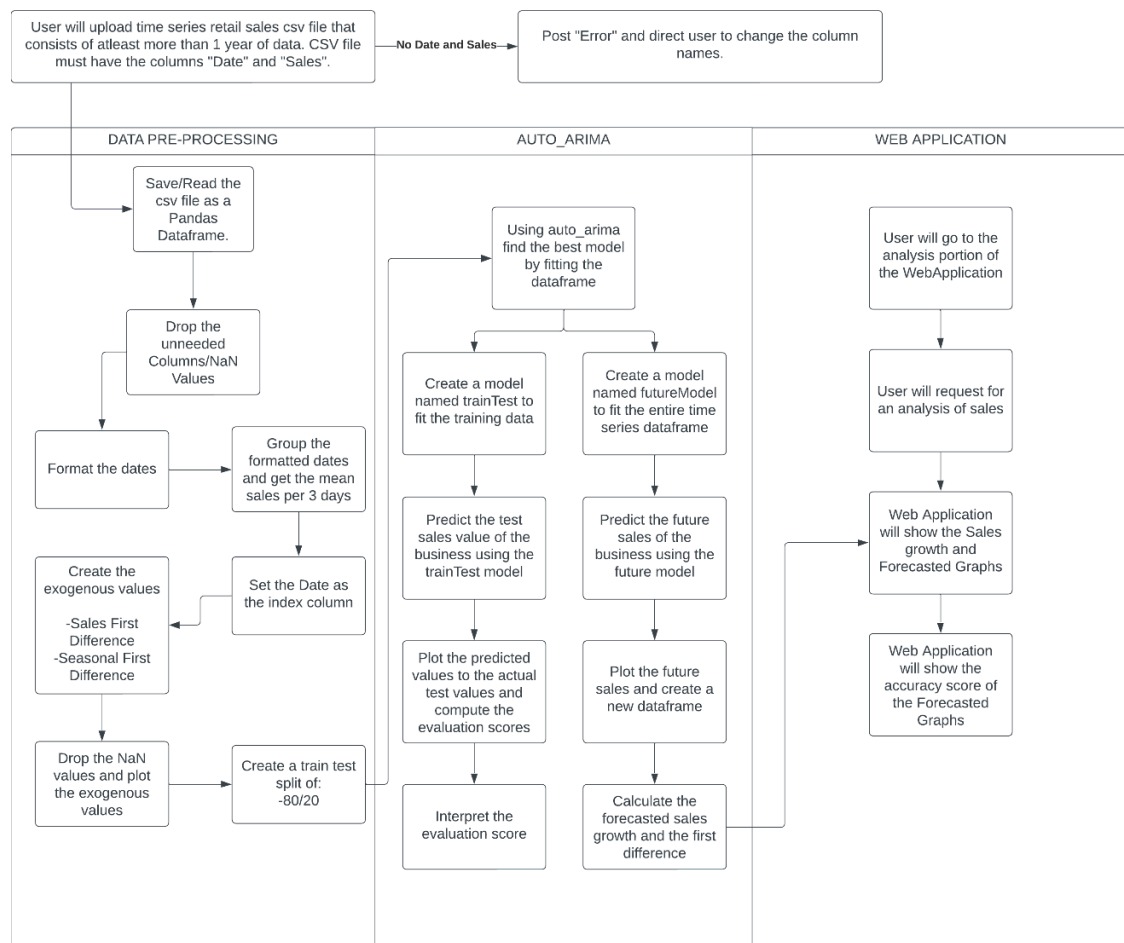


Figure 3.4.1 shows the Operational Framework of the system. It specifically shows the process of the system during the sales analysis portion. The system starts when the user



uploads their time series data, and when they press the FORECAST button. The system may also use data from the inventory system like past sales. After this, the system would then go into the pre-process phase where the data would be prepared and cleaned for the next phase.

The pre-process phase mainly includes grouping the data and cleaning of NaN values. The next phase would be the fitting phase where the cleaned data would be used to fit the SARIMAX model using `auto_arima`. The model would then be used to predict future sales and sales growth. It is also used to calculate the evaluation or accuracy metrics of the prediction vs the actual test or sales data.

After this phase the next phase of the system involves the actual Smart Inventory Systems and is only accessible for admins of the company. This phase starts with the user navigating towards the Inventory page. Once there, the admin can either add, edit, or delete entries. Lastly, the system will output the generated invoice as an Excel file and the user will then give their feedback on the model.



3.5 USE CASES

Figure 3.5.1.

Use Case Diagram of the System.

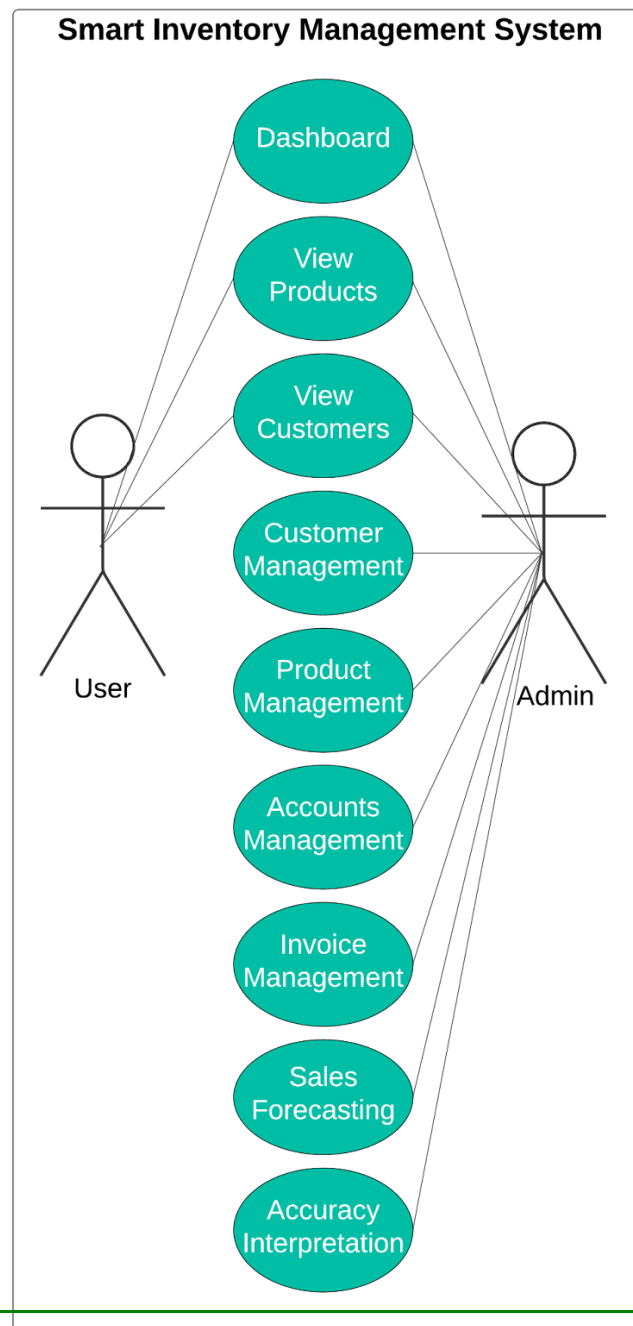




Figure 3.5.1 encompasses the use cases for the user, admin, and the system. Based on the needs of the company and the people who will use it. There are a total of two types of accounts, which are the user and admin accounts. The user accounts will only be able to view the dashboard and the customer and products table. The admins on the other hand will be able to have more access specifically to the actual inventory management, sales analysis, and archiving of previous data. Figure 3.5.1 also shows the relationship of these two types of accounts based on the use cases.

3.6 RESPONDENTS OF THE STUDY

Emdee Chem Marketing Corporation has multiple departments such as Warehousing, Production, and Marketing. As such these three departments are the focus of this study. However, they also have other departments such as sales, logistics, technical, and quality. Their main department is located at their main office in Quezon City in 16, Dividend Street, GSIS Village, Project 8, Quezon City, 1106 Metro Manila. and has currently fifteen employees.

Each department is handled by an admin who oversees the operations and regular employees who work under the admins. The estimated number of respondents for this study will be around five admins and ten workers if their schedules permit it.



CHAPTER IV

This section of the study will go into detail on the design and use of each module of the proposed system. Additionally, the methodology will be discussed in this module. data gathering and analysis. The web application was coded in Visual Studio Code and is deployed online through LightNode.

4.1. DESIGN AND METHODOLOGY

The research employs a descriptive and applied approach. The study is descriptive in how it focuses on analyzing the existing sales data to forecast future trends and make sense of existing sales patterns. Additionally, the study is applied as it solves problems for the company and develops a system that addresses these problems. These designs were chosen because they best suit the study and how they shape the data collection and analysis.

The study employed the questionnaire type of method to evaluate the system. Using well-structured questionnaires with the employees of the company who will be using the system and admins who have experience working with the traditional log booking system. The researchers were able to gather the company's evaluation of the two systems. The sampling method for the data gathering was convenience sampling as per the advice of the statistician of the study, given that the population of the relevant employees were limited.



After gathering the results of the evaluation, the data was treated by the statistician using descriptive statistics and a t-test was made to compare the evaluation of the traditional logbook system and the proposed smart inventory management system.

For the ethical considerations of the study, the researchers plan to give the respondents an Informed Consent Form (ICF) in their preferred language of either English or Tagalog. The main concern the researchers and the beneficiary had with the ethics of the study is the ethical use of their data. The researchers focused on making sure that the sales data provided by the company would be treated with the highest ethical considerations. The group solved this consideration by only using the data to test the model. Any trace of actual sales data will not be included in any documentation, instead the company has advised the users to use dummy data and only use publicly available data such as the company website. See Appendix H for more details.

Additionally, the ethics department of the university checked and approved the study and gave the researchers an ethics certificate last November 4, 2024, certifying that the committee approved the research. See Appendix D for the signed ethics certificate.



4.1.1. Application Introduction

Figure 4.1.1.1

Emdee SIMS Home Screen

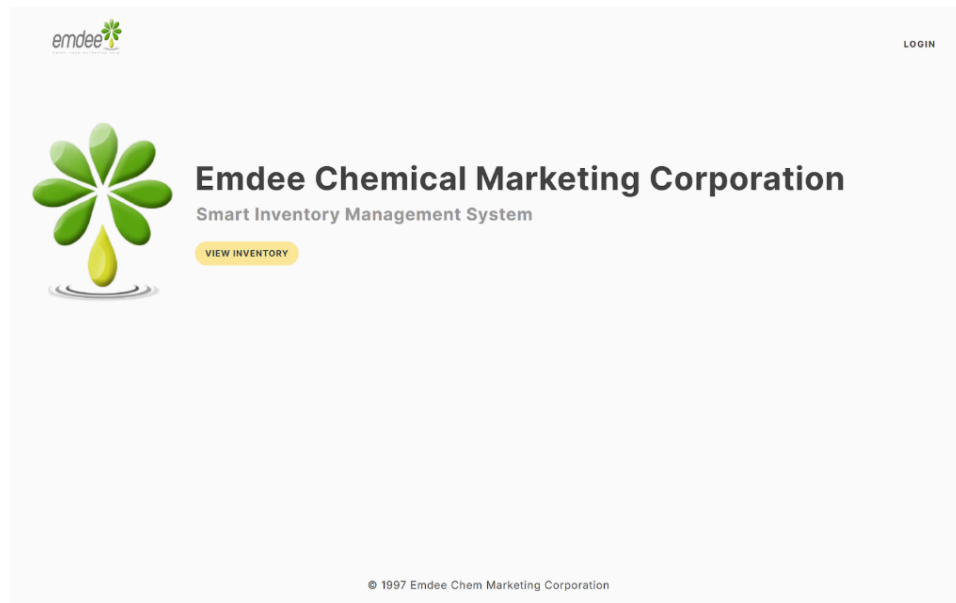


Figure 4.1.1.2

Emdee SIMS Log-in

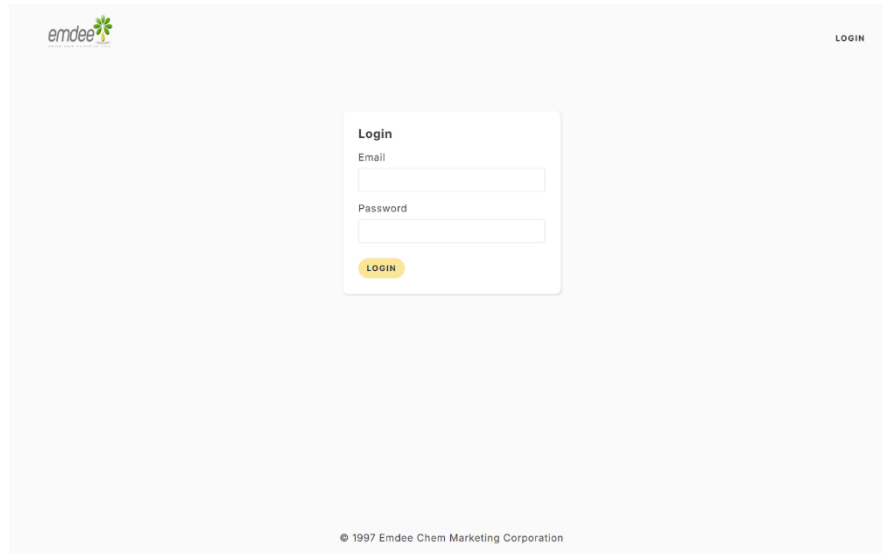
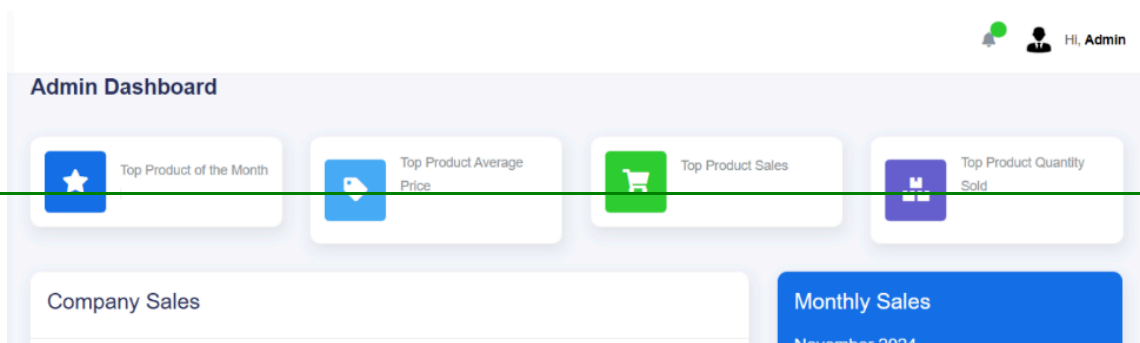


Figure 4.1.1.1 showcases the landing page wherein the user of the web application is greeted with the company's logo, name, and is prompted to either log-in or view the inventory. If the user were to click either of the two, the user is then redirected to Figure 4.1.1.2 for the login view of the system. This log-in page accepts both user and admin login given that they can provide the necessary credentials. Once logged in to the web application, the user will be redirected to the dashboard view. The dashboard view is dependent on the type of account that is currently logged in, which is either the admin or user view.

Figure 4.1.1.3

Emdee SIMS Admin Dashboard





**Figure 4.1.1.4***Emdee SIMS Admin Recent Invoices*

Recent Invoices			
INVOICE FILE	INVOICE DATE	CUSTOMER NAME	TOTAL PRICE
	2024-05-03		
	2024-04-30		
	2024-04-30		
	2024-04-30		
	2024-04-29		
	2024-04-26		
	2024-04-26		

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Figure 4.1.1.3 and Figure 4.1.1.4 showcase what the admin will see once the login is successful. The different features of the dashboard include a sidebar for quick access of the different views of the application, a navigation bar wherein the user can check for notifications and click on the profile to log out.

The dashboard for both user and admin contain specific data based on the top product of the month, this includes the name, price, quantity sold, and sales of the top product. There is a graph that displays the sales of the company for the year and beside it is the recent month and its sales as well as a section that notifies the stock to the user.



The only difference between the user and the admin dashboard is that the admin dashboard can see the recent invoices underneath the main content. This is shown in Figure 4.1.1.4.

**Figure 4.1.1.5***Emdee SIMS User Customer Table*

Emdee SIMS

Dashboard Inventory Customer Product

User Customer

Show 10 entries

Search

CUSTOMER NAME	CUSTOMER TYPE
	Company
	Company
	Company
	Company
	Company
	Individual
	Company
	Company
	Company
	Company

Customer Name

Customer Type

Showing 1 to 10 of 156 entries

Previous 1 2 3 4 5 ... 16 Next

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Figure 4.1.1.5. showcases the user customer table and it only contains the columns for customer name and customer type, and the table is for view only. Additionally, the user sidebar only contains views for the customer table and product table, significantly less than what the admin sidebar has.

**Figure 4.1.1.6***Emdee SIMS Admin Customer Table*

Emdee SIMS

Admin Customer

Show 10 entries

Search

CUSTOMER NAME	CUSTOMER TYPE	EMAIL	ADDRESS	ACTION
AMENCAR	Individual	juliancamadon19cin@gmail.com	Tagaytay	Edit Delete
	Company	no email	Quezon City	Edit Delete
	Company	qwert@mail.com	Makati	Edit Delete
	Company			Edit Delete
	Company			Edit Delete
	Company			Edit Delete
	Individual			Edit Delete
	Company			Edit Delete
	Company			Edit Delete
	Company			Edit Delete

CUSTOMER NAME CUSTOMER TYPE ADDRESS EMAIL ACTION

Figure 4.1.1.6 showcases what the admin would view if they were to check the customer table. Notable differences from the user customer table are the inclusion of two additional columns, email and address. As specified by the beneficiary that they only wish to divulge the email and address information to the admins of the company as they are the ones who will contact the customers and plan the deliveries. Another different is the inclusion of an Add Row button to add a new customer, edit button to edit the details of an existing customer, and delete button to remove a customer from the database (see Appendix B for the screenshot of the add customer modal and the different alerts of success and failure for the add, edit, and delete).

**Figure 4.1.1.7**

Emdee SIMS Admin Product Table

Admin Product

+ Add Product

Show 10 entries

Search:

PRODUCT NAME	PRODUCT TYPE	STOCK	THRESHOLD	SIZE	LAST BOUGHT	ACTION

Figure 4.1.1.7 showcases what the admin would view if they were to check the product table. Both user and admin product table share the same columns of product name, type, stock, and description. However, only the admin can add new products to the database, edit existing products, and delete products.

The features from here on out are only exclusive to the admins as the users can only access the dashboard and view the product and customer tables.



**Figure 4.1.1.8***Emdee SIMS Admin Accounts Table*

User Accounts Table [Add Account](#)

Show entries Search:

FIRST NAME	MIDDLE NAME	LAST NAME	SUFFIX	MOBILE NUMBER	EMAIL	ACTION
CEDRIC	C	NEDTRAN				✕ ✕
CHARLES	D	VILLANUEVA				✕ ✕
JULIAN	A	ENCARNACION			julianencarnacion19slm@gmail.com	✕ ✕
MARIA	D	DOE			user@gmail.com	✕ ✕

Showing 1 to 4 of 4 entries [Previous](#) [1](#) [Next](#)

Figure 4.1.1.9*Emdee SIMS Create Account View*

Create Account [View Accounts](#)

Account Details

First Name

Middle Name

Last Name

Suffix

Mobile Number

Email Address

Password

At least 8 characters

Confirm Password

[Create Account](#)



Figure 4.1.1.8 showcases what the admin would see if they were to access the accounts, and it details the information of the user accounts that can access the inventory system. The admin also has the power to edit the account details, delete the account, and create new user accounts. There is a button to create accounts, and it redirects the admin from the table into the create account view.

Figure 4.1.1.9. showcases what will be the admin will see if he or she wishes to create a new user account. The admin should fill up all the fields except the suffix bar and once all of the inputs are validated to have the correct values there will be a pop-up to notify the user that the account has been created.

Figure 4.1.1.10

Emdee SIMS Admin Invoice

FILE NAME	INVOICE DATE	CUSTOMER NAME	DISCOUNT	TOTAL PRICE	DOWNLOAD	ACTION
appendix1	2024-12-10	JANE DOE	10	90		
	2024-05-03		0			
	2024-04-30		0			
	2024-04-30		0			
	2024-04-30		0			
	2024-04-29		0			
	2024-04-26		0			
	2024-04-26		0			
	2024-04-26		0			

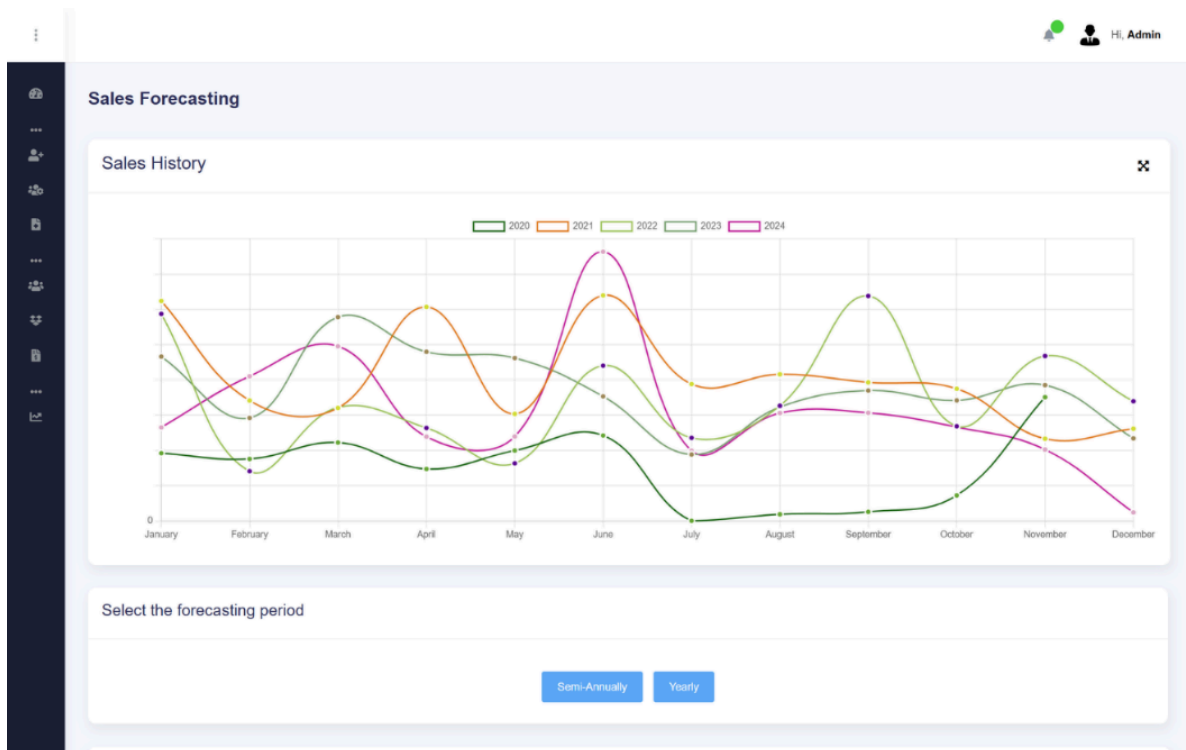


Figure 4.1.1.10 showcases the view of the admin invoice. In this table it displays the file name of the invoice, date of the invoice, customer name, discount percent, and total price. The view also has a button to redirect the admin from the table of invoices to the create invoice view.

The table has an additional button for the automatic invoice generation and the edit invoice and delete invoice function. (see Appendix G for the example of the creation of the invoice and its automated downloaded invoice as an excel file).

Figure 4.1.1.11

Emdee SIMS Sales Forecasting Details



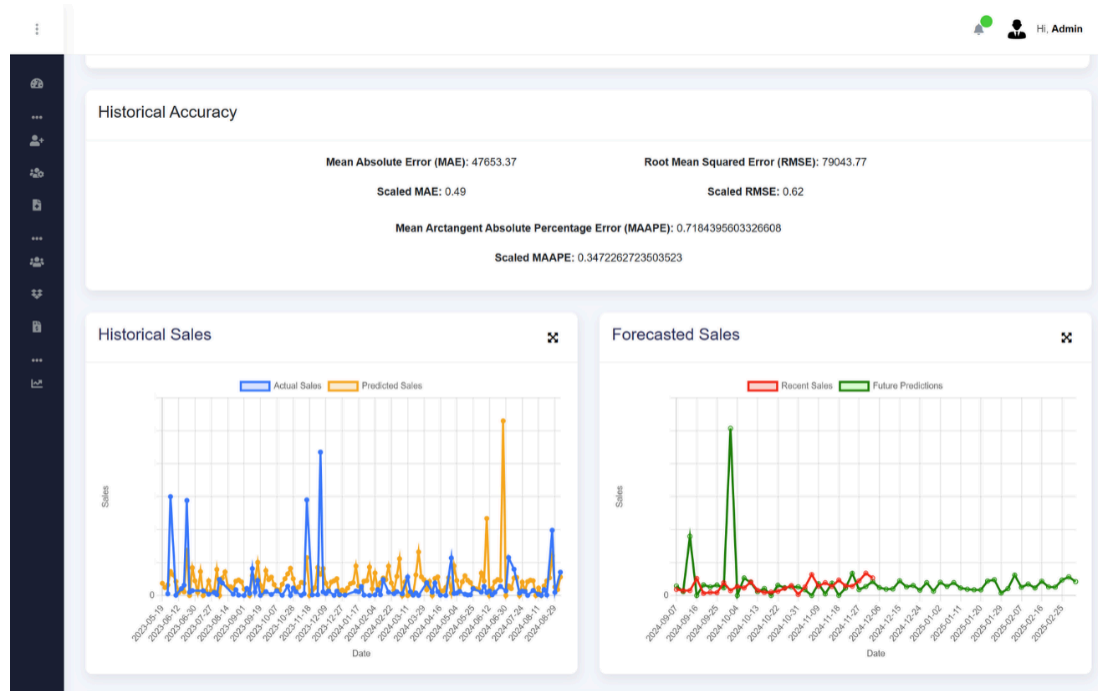
**Figure 4.1.1.12***Emdee SIMS Sales Forecasting Output*

Figure 4.1.1.11 and Figure 4.1.1.12 both showcase the input and output of the Sales Forecasting, only exclusive for the admins of the company. The details of the Sales Forecasting have a graph to visualize all of the sales data that will be used in the forecasting and asks the admin the period for the forecast and to click the start forecast to begin. Once the forecast is completed the accuracy of the forecast is displayed in MAPE, MAE, and RMSE. There will also be two outputted graphs one for the historical sales to compare the actual sales data with the predicted sales data, beside it is the forecasted sales for the future values and the table below has the specifics of the future forecasted sales data. Additionally, all of the graphs in this module have a full screen function for viewing the results better.



4.2. PROJECT DEVELOPMENT

The proponents of the research followed the discipline and suggested measures of Rapid Application Development (RAD). As the development focused on quickly developing a working prototype through iterative cycles of design and build. The requirements from the beneficiary were defined early and their feedback was implemented during the prototype phase.

The researchers implemented the requirements planning phase by first consulting the beneficiary to ask about their needs and wants in the system. Additionally, the researchers also found references on the common modules that an inventory management system would have. The second step in the development process, which is User Design, was done by first developing a prototype and presenting it to the proponent for initial review. Feedback was then given and taken into account when creating the actual system itself. The user stories and scenarios are also decided and finalized in this stage. Lastly, the testing of the model and checking of the accuracy is being done in this stage.

During the Construction stage the researchers created wireframes for the front-end designs of the system. During the construction of the front end design the researchers coordinated with the proponent to further create their desired frontend functionalities and design using EmbeddedJavaScript (EJS). It was also used to create



the HTML pages itself, while the styling was made with the assistance of a CSS framework called Bootstrap. Lastly, to visualize numerical data, ChartJS is used for seamless integration between the JSON data from the backend and the Machine Learning API. For the backend, the researchers made use of a SQL database using PostgreSQL. Additionally, JSONWebTokens and Node-Express Framework were used for authentication, and server-side operations respectively. Considering that the system would be handling sensitive information such as invoices and sales data, BCrypt is implemented to add another layer of protection as necessary authentication to access data being stored. Lastly, DrizzleORM is used to allow queries from TypeScript to communicate with the SQL database. For the ML API module, it is mainly developed using FastAPI as its framework. The packages that were used for the sales prediction were pmdarima and statsmodels.

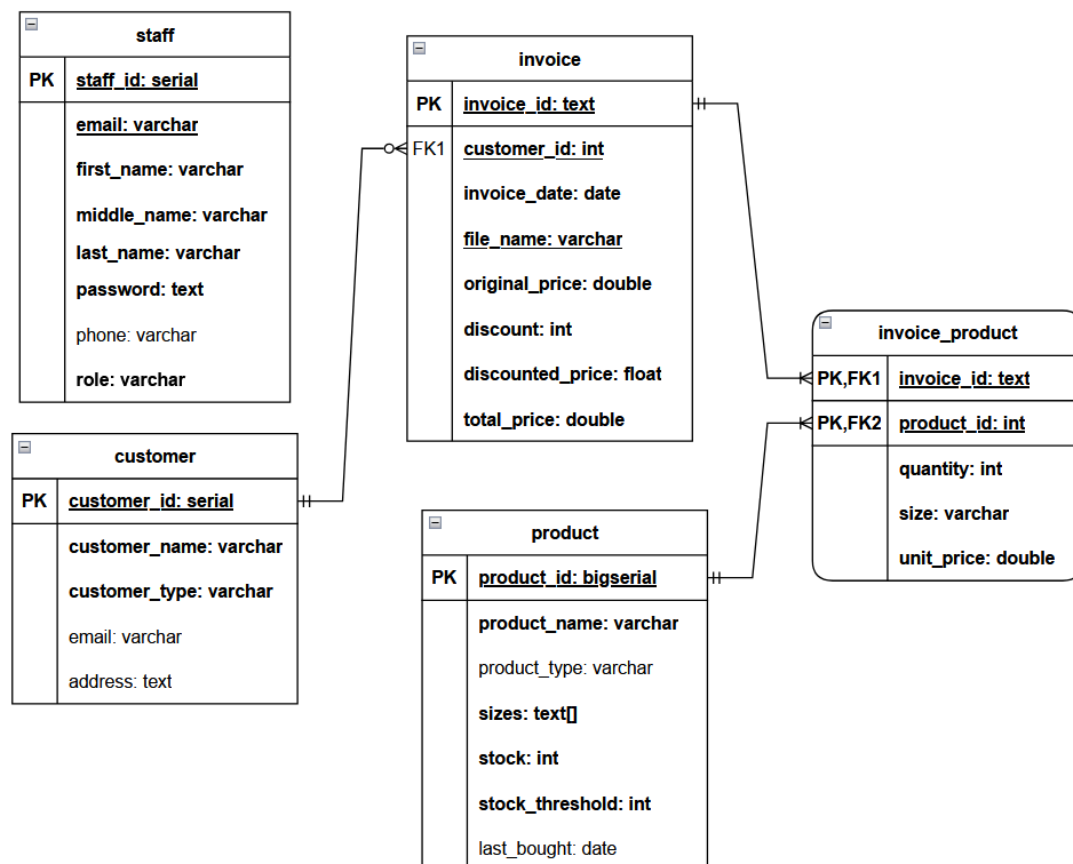
Specifically Figure 4.2.1 details the database schema model of the inventory system ideal for the company's requirements with four interconnected tables, namely: customer, invoice, product, and invoice-product. The customer table stores information about customers and connects to invoices via a foreign key (customer id), linking invoices to their respective customers. The invoice table records general invoice details (e.g., date, pricing, and discounts) and connects to invoice-product through invoice id. The product table contains details about available products and connects to invoice-product through the product id. The invoice-product table acts as a



junction table, associating multiple products with a single invoice, and includes additional fields like quantity, unit price, and size for product-specific details within an invoice. Lastly, the staff table stores user information such as personal information, email and password for authentication, and user roles such as admin and user.

Figure 4.2.1.

Database Schema



During the Cutover phase, the system was deployed on a Virtual Private Server (VPS) provided by LightNode. A firewall was configured using UFW, and NGINX



was set up as a reverse proxy to secure and manage incoming internet traffic to the server. Initially, access was restricted to the researchers. However, the domain 'emdee-sims.online' is configured via GoDaddy to route traffic to the server's address for the employees' access. After a successful deployment, the researchers then visited the company's office to guide employees in testing the system and gathering valuable feedback.

4.3. DEVELOPMENT PLANNING

4.3.1. Software Specifications

The different software used for building and developing the web application are as follows:

1. Visual Studio Code also known as VS Code. This is a code editor that allows for a multitude of extensions that allow for help with coding, smarter comments and suggestions, and built-in support for Typescript and Node.js. VS Code offers the following features that help in the development of the application:

- Integration with GitHub Copilot for code completion and context awareness
- Ease of integration with Virtual Environment
- Ease of terminal usage specifically for debugging, output checking, and local server testing



2. Google Colab is used for the Machine Learning module. Google Colab allows the hosting of Python notebooks and free access to computing resources, such as GPUs and TPUs. Google Colab was also picked over Jupyter because of the ease of accessing the file on different devices since it is saved on the cloud. Additionally, running the fitting on Google Colab would more closely simulate how the API would handle fitting with limited resources.

3. DBeaver is used for the database section as it is a multifunctional SQL query tool for developing, storing, and administering databases. It ideally supports PostgreSQL and various other database management systems and provides an intuitive interface for writing and executing SQL queries, visualizing database schemas, and managing data effectively. It also aids in ensuring the database aligns with the system's requirements.

4. GitHub Desktop is used to simplify the development workflow. GitHub Desktop is an open-source application that allows users to work with code hosted on GitHub and through the implemented GUI, it simplifies the process of pushing and pulling changes.

5. Postman is used for testing and developing backend APIs, its features include request building and response validation. It simplifies collaboration by allowing the group to share API test collections. In testing the inventory management



system, it can streamline testing, validate CRUD operations, and ensure seamless communication between the backend and frontend.



4.3.2. Language Used

These were the different programming languages used in the making of the web application.

1. Python is a high-level programming language used for the development of the Machine Learning API and implementation of the SARIMAX model. The different Python libraries allow the importing of different machine-learning models, and the manipulation of data easily using pandas. FastAPI is also used to allow the Python code to be turned into a working API that returns json values to be used by both the backend and the front end. Additionally, the integration of the openpyxl library has made it possible for invoice templating and generation.

2. TypeScript is a modified version of JavaScript which is the primary language in the framework of NodeJS/Express. It allows for tighter integration and type-safety for developing backend functionality and is simple to link with the relational database using object-relational mapping libraries such as DrizzleORM.

3. PostgreSQL is crucial to the inventory management system and will be the primary database for storing and managing inventory-related data and related information. Its features such as support for forming complex queries and ACID compliance, make it an ideal choice for ensuring data integrity and reliability.



4.4. EVALUATION OF THE PROJECT

4.4.1. Respondents of the Study

There was a total of six respondents during the data gathering period, all consisting of admins of the company. The admins were all from different departments from the Staff Admin, Marketing and Production Admin, Technical and Quality Admin, Production Supervisor, Sales Admin, and Production Admin.

4.4.2. Research Instruments and Techniques

The researchers employed the use of survey questionnaires to obtain the primary data for the evaluation of the system. The survey questionnaires were printed and given to the company respondents along with the Informed Consent Forms for ethical considerations. The quantitative answers from the evaluation were all collated inside an Excel file and were sent to the research's statistician for the descriptive statistics as well as the t-test for the comparison of the results from the logbook system and the smart system. The statistician used Jamovi and Comprehensive R (CRAN) for the statistical treatment of the results.

4.4.3. Data Gathering



The researchers of the study gathered information that provided the data for the results and discussion of the study. The information gathered through the survey serves as the evaluation of the traditional logbook system and the smart inventory system. Additionally, there were suggestions and comments from the respondents that detail their experiences with the traditional logbook system, their level of inventory accuracy of both traditional and the proposed system, and suggestions to improve the application, which is discussed in the recommendation section.

4.4.4. Questionnaires and Surveys

Survey questionnaires were employed as the primary data-gathering instrument for the evaluation of the web application by the employees and admins of Emdee Chemical Marketing Corporation. The survey included a likert scale scored from 4 up until 1 with 4 being Good and 1 being Needs Improvement. This scale is seen in Figure 4.4.3.1. The surveys were approved by the faculty of De La Salle University Dasmariñas and the signed copies of both traditional system survey and the proposed survey can be seen in Appendix A.

Figure 4.4.4.1.

Evaluation Scale

SCALE:



4 - Good	3 - Satisfactory	2 - Below Average	1 - Needs Improvement
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The questionnaire categories were mainly based on the terms and definitions found in the ISO/IEC 25010:2023 document. This document is mainly used for Systems and software quality evaluation and requirements creation. The categories used were Functional Completeness, Functional Correctness, Time behaviour, Operability, Security, and Scalability. Functional Completeness and Correctness is used to describe if the functions that fit the intended objectives are covered, and if the functions are able to provide accurate results. Time behaviour is used to describe the capability of the system to be able to perform in a timely manner. Operability describes the characteristics and functions of the system that allow it to be simple to operate. Security discusses the ability of the system to protect sensitive data that is present in the database or the system itself. Lastly, Scalability discusses the ability of the product to handle different workloads, and the potential of the system in the future.



CHAPTER V

RESULTS AND DISCUSSION

5.1 MODEL EVALUATION

Type of Data	SARIMAX	MAE	RMSE	MAAPE	Scaled	Scaled	Scaled
	Model Used	Score	Score	Score	MAE	RMSE	MAAPE
					Score	Score	Score
Train/Test	SARIMAX	47653.37	79043.77	0.7184	0.49	0.62	0.3472
(70/30 split)	(0, 0, 1) x (2, 1, 0, 12)						
Train/Test	SARIMAX	33312.86	82771.01	0.5451	0.27	0.33	0.2078
(80/20 split)	(0, 0, 1) x (2, 1, 0, 12)						
Future	SARIMAX	36053.77	46349.53	0.5662	2.58	3.31	0.5179
	(0, 0, 1) x (2, 1, 0, 12)						

Table 5.1.1

Table of models used and the accuracy scores each of the models received

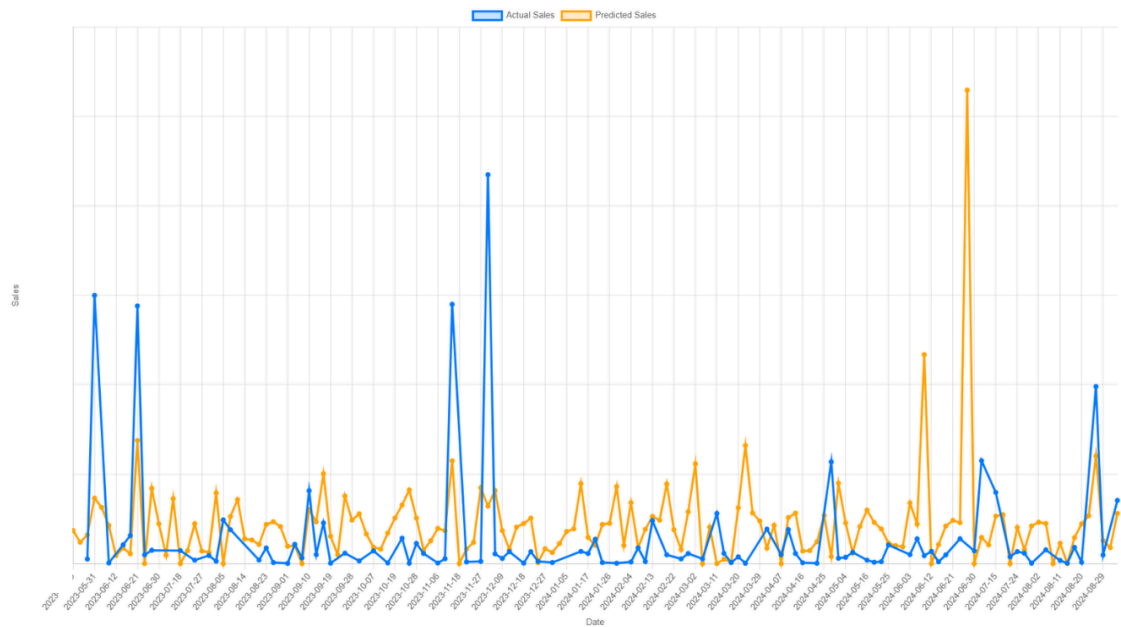
MAE, RMSE, and MAAPE is used to calculate the accuracy of the prediction because MAE and RMSE are scale dependent metrics typically used in ML accuracy evaluation. MAAPE is used over MAPE due to it being a scale independent metric. Table 5.1.1 showcased the different models used by the different train and test splits



and the accuracy scores of each type of data. There are two categories overall for accuracy, non-scaled and scaled values. The scaling is done by using MinMaxScaler from sklearn. Scaling is done to be able to scale the values from thousands into a more readable metric. Additionally, one of the reasons why it is scaled down is because since RMSE and MAE are scale dependent metrics the sheer size of the values affect these accuracy values. By scaling them down and getting the RMSE and MAE score it is easier to understand the error. This is also done since there are data points that are measured in thousands and millions, comparing the two scales would greatly affect the accuracy score. Once scaled down the split that performed the best would be the 80/20 split with a 0.27 MAE and 0.33 RMSE score. Even though MAAPE is a scale independent measure, scaling is still performed to see if it had any significant changes. The model that had the lowest MAAPE or absolute percentage error would be the 80/20 split, namely having a 54.51% and 20.78% error for the non-scaled and scaled accuracy scores. As for the models used, each type of data had its model fitted due to auto_arima performing fitwise to detect the best model to be used with means of detecting which model had the least amount of AIC. The accuracy scores for the future predictions are gathered by getting the 30 most recent periods and using those periods to verify the predictions.

**Figure 5.1.1**

Graph of the Original Sales data vs the Predicted sales using a 70/30 train test split



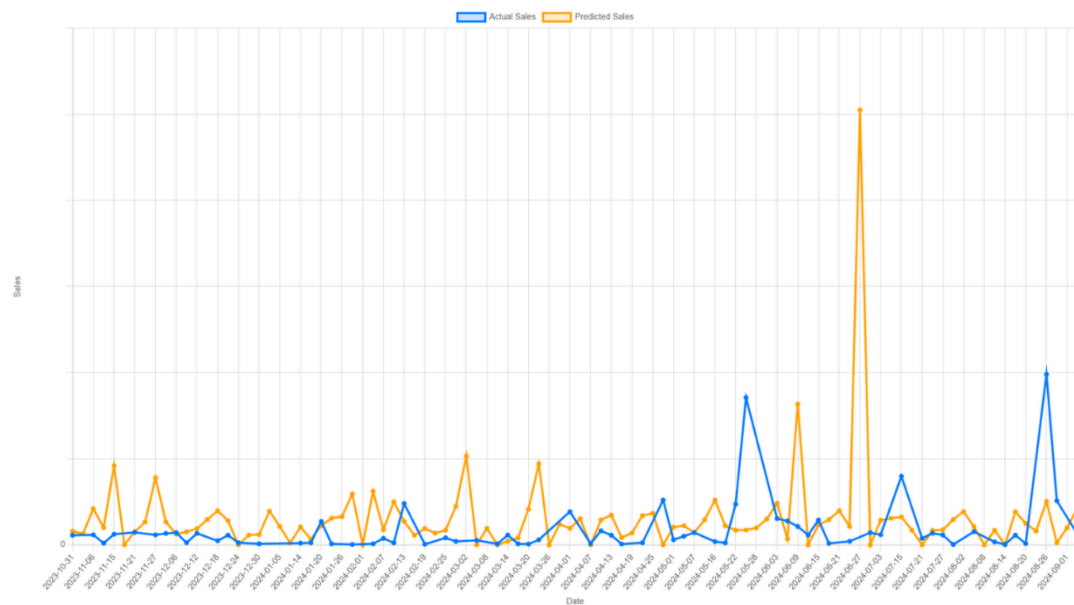
Figures 5.1.1 and 5.1.2 showcase the sales data compared to the test prediction. Due to the beneficiary not consenting to the display of the sales value and dates the researchers were able to negotiate the use of the graph without the indexes on the x and y axis. Looking at the trendline of the predicted sales compared to the actual sales it could be observed that the model was able to predict the sudden rises of sales which was one of the indicators for bulk manufacturing and buying. It was also able to predict the trend during days when sales suddenly dropped or were low. However,



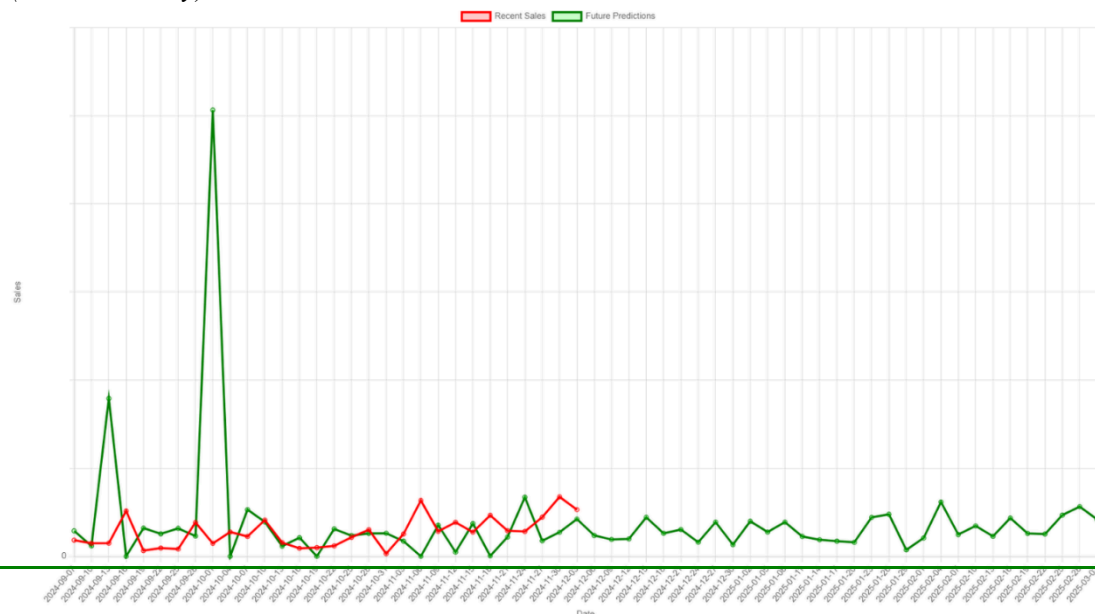
during days with really high manufacturing and sales the model predicts values that go into the negative.

**Figure 5.1.2**

Graph of the Original Sales data vs the Predicted sales using a 80/20 train test split

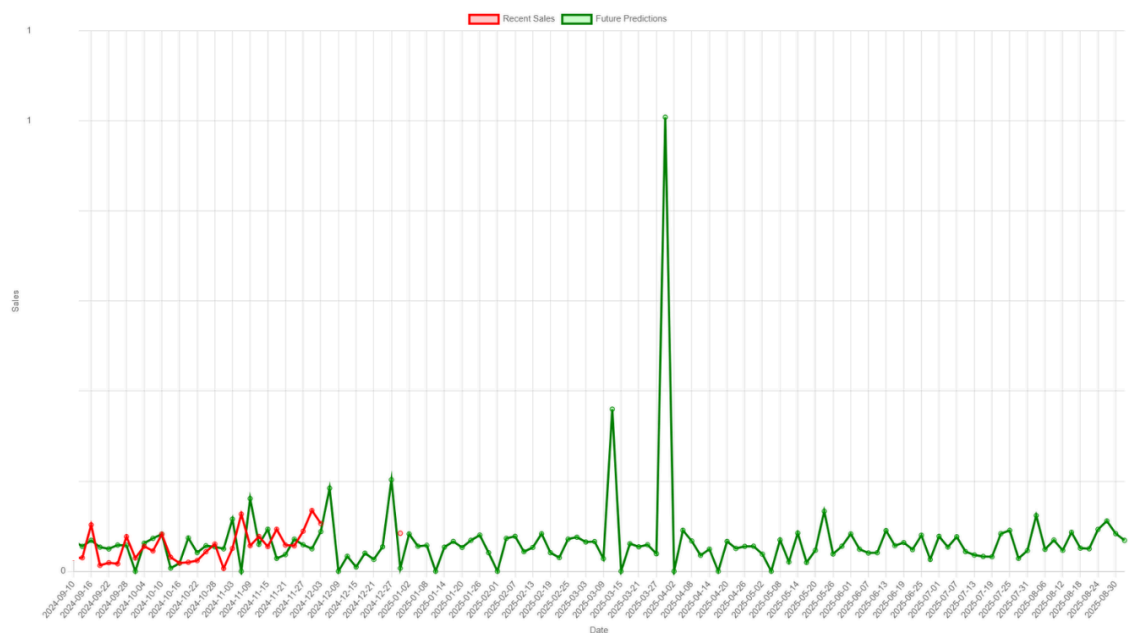
**Figure 5.1.3**

Graph of the last 30 periods of sales compared to the future sales (SemiAnnually)



**Figure 5.1.4**

Graph of the last 30 periods of sales compared to the future sales (Annually)



Figures 5.1.3 and 5.1.4 showcase the split that was done to compare the accuracy of the future prediction. The last 30 periods or 3 months were used to get the accuracy of the future on those periods. Furthermore, the Future Prediction line can predict from 6 months until a year of data. The future prediction itself is also dependent on how many $n_periods$ or days were used in the prediction.





5.2 SURVEY EVALUATION

Figure 5.2.1

Graph of the Respondents evaluation on the Functional Correctness of the Traditional System

Functional Correctness - Traditional System

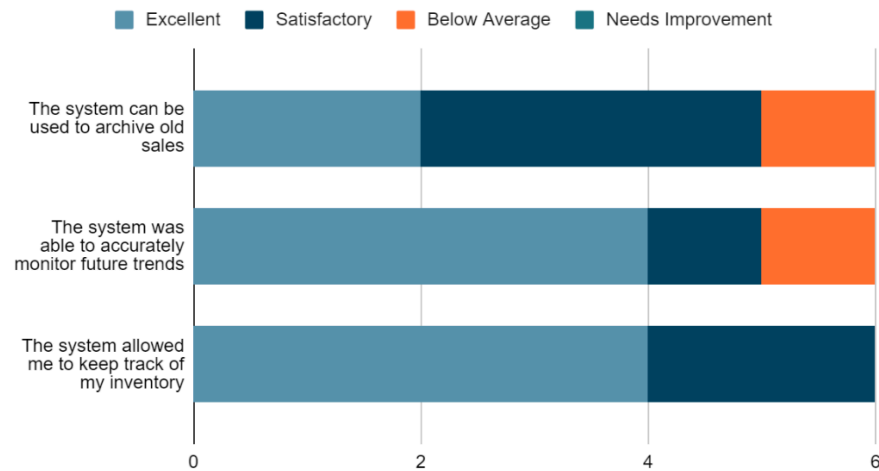
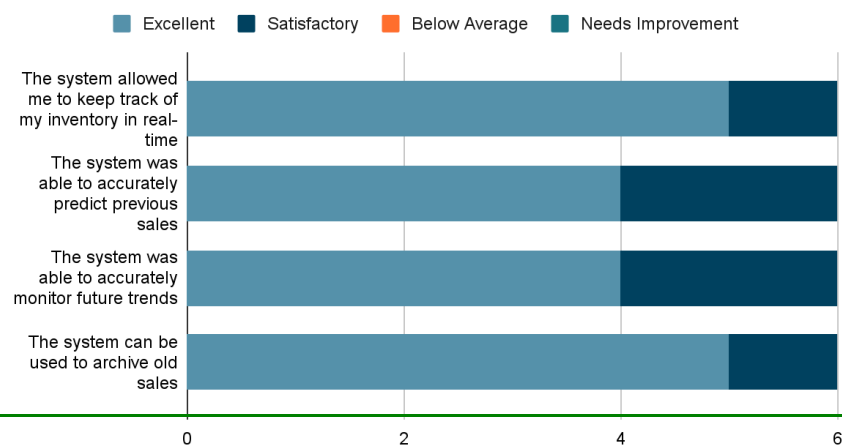


Figure 5.2.2

Graph of the Respondents evaluation on the Functional Correctness of the Smart Inventory System

Functional Correctness - Smart Inventory





Figures 5.2.1 & 5.2.2 showed the respondents evaluation on the functional correctness of both of the systems. The respondents had an average of 3.389 for the entire category on the traditional system, which is an average interpretation of Satisfactory for the system's Functional Correctness. Meanwhile, for the smart inventory system, the respondents had an average of 3.722 or an interpretation of Excellent. Based on the rounded averages of the respondents, it can be observed that the respondents preferred the smart inventory system over the traditional system mainly with a rounded-up score of 4 to 3 or Excellent to Satisfactory in favor to the smart inventory system.

Figure 5.2.3

Graph of the Respondents evaluation on the Operability of the Traditional System

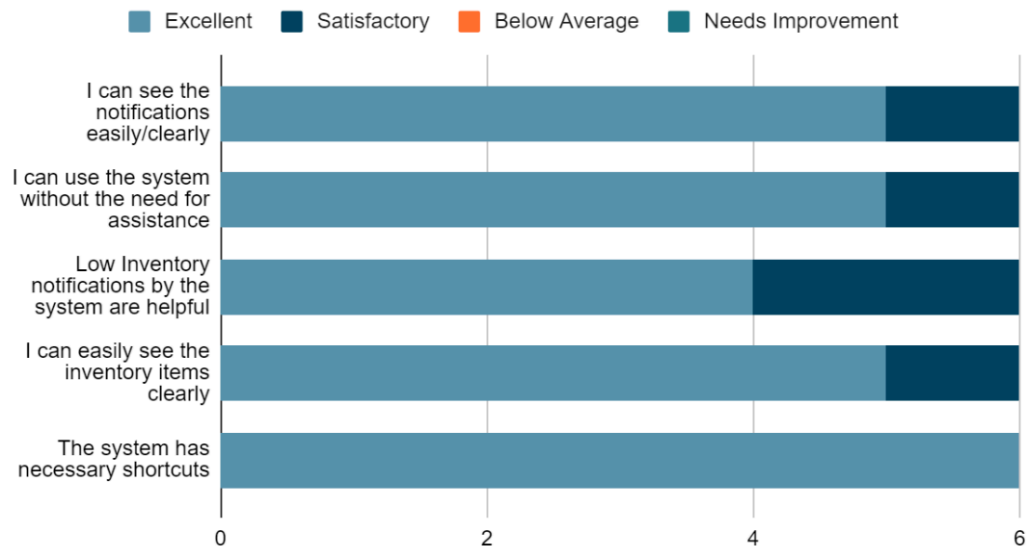
Operability - Traditional System



**Figure 5.2.4**

Graph of the Respondents evaluation on the Operability of the Smart Inventory System

Operability - Smart Inventory



Figures 5.2.3 & 5.2.4 showed the respondents evaluation on the Operability of both of the systems. The respondents had an average of 2.388 for the entire category on the traditional system, which is an average interpretation of Below Average for the system's Operability. Meanwhile, for the smart inventory system, the respondents had an average of 3.833 or an interpretation of Excellent. Based on the rounded averages of the respondents, it can be observed that the respondents preferred the smart inventory system over the



traditional system mainly with a rounded-up score of 4 to 2 or Excellent to Below Average in favor of the smart inventory system.

Figure 5.2.5

Graph of the Respondents evaluation on the Security of the Traditional System

Security - Traditional System

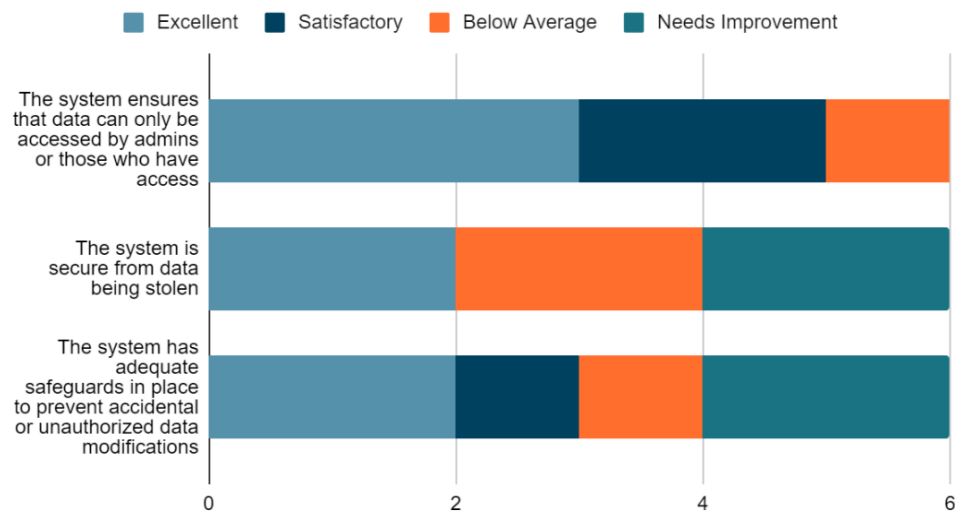
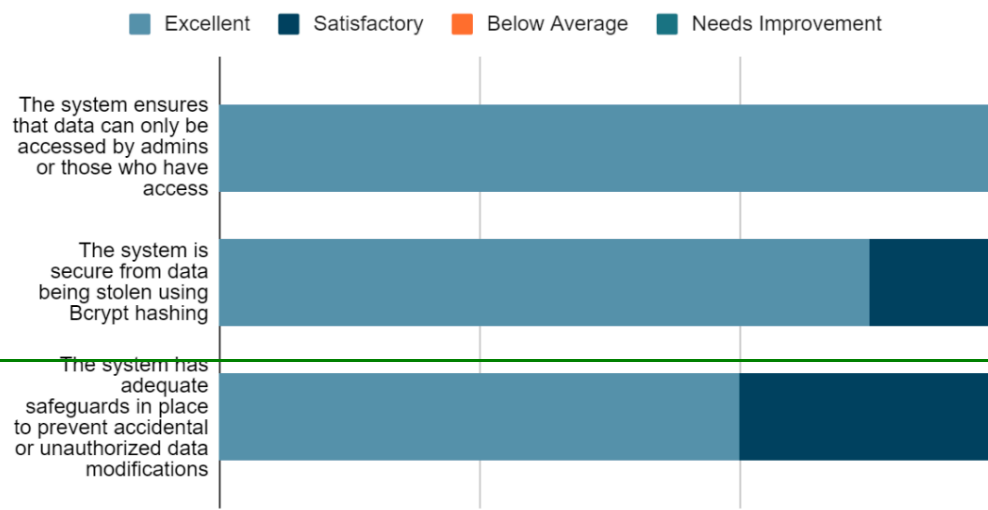


Figure 5.2.6

Graph of the Respondents evaluation on the Security of the Smart Inventory System

Security - Smart Inventory





Figures 5.2.5 & 5.2.6 showed the respondents evaluation on the Security of both of the systems. The respondents had an average of 2.722 for the entire category on the traditional system, which is an average interpretation of Satisfactory for the system's Security. Meanwhile, for the smart inventory system, the respondents had an average of 3.833 or an interpretation of Excellent. Based on the rounded averages of the respondents, it can be observed that the respondents preferred the smart inventory system over the traditional system mainly with a rounded-up score of 4 to 3 or Excellent to Satisfactory in favor of the smart inventory system.

Figure 5.2.7

Graph of the Respondents' evaluation on the Scalability of the Traditional System



Scalability - Traditional System

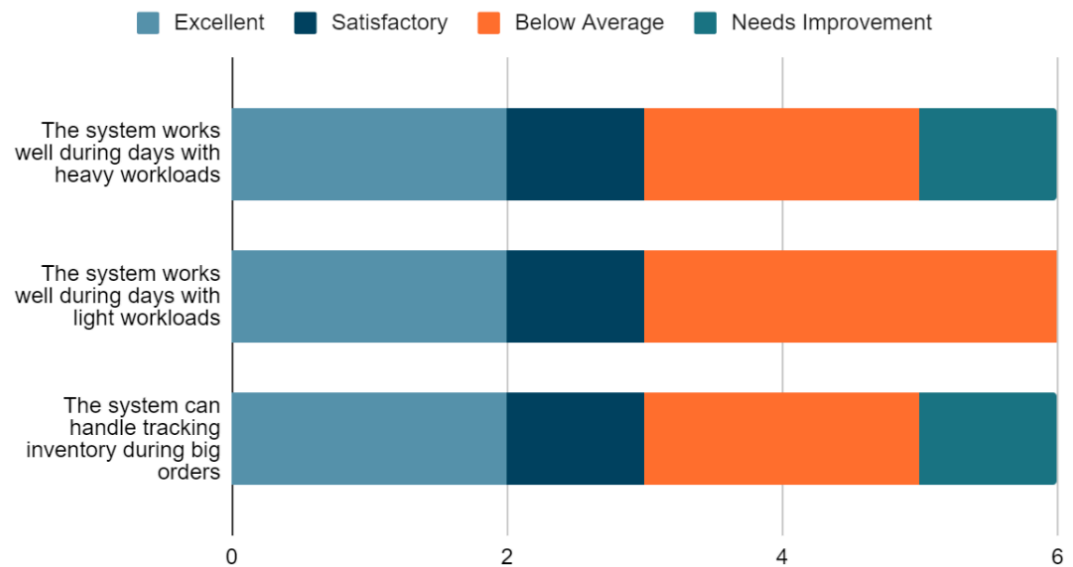
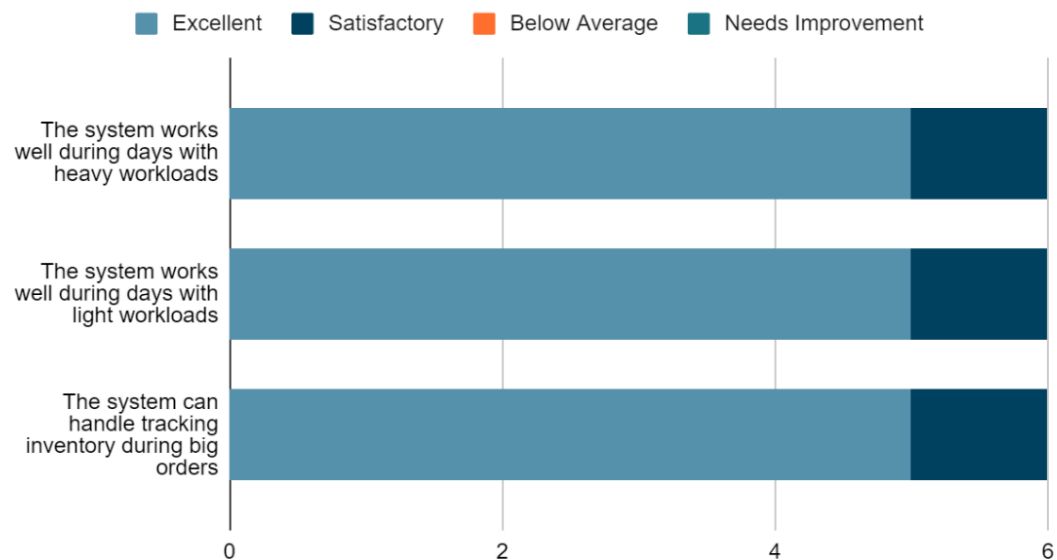


Figure 5.2.8

Graph of the Respondents evaluation on the Scalability of the Smart Inventory System



Scalability - Smart Inventory



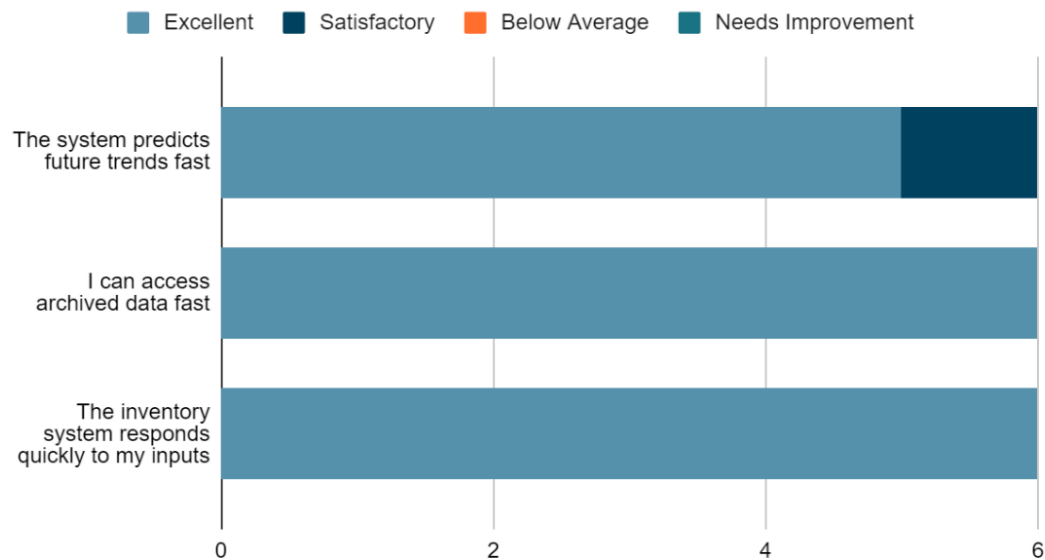
Figures 5.2.7 & 5.2.8 showed the respondents evaluation on the Scalability of both systems. The respondents had an average of 2.722 for the entire category on the traditional system, which is an average interpretation of Satisfactory for the system's Scalability. Meanwhile, for the smart inventory system, the respondents had an average of 3.833 or an interpretation of Excellent. Based on the rounded averages of the respondents, it can be observed that the respondents preferred the smart inventory system over the traditional system mainly with a rounded-up score of 4 to 3 or Excellent to Satisfactory in favor to the smart inventory system.

Figure 5.2.9



Graph of the Respondents evaluation on the Time Behaviour of the Smart Inventory System

Time Behaviour - Smart Inventory

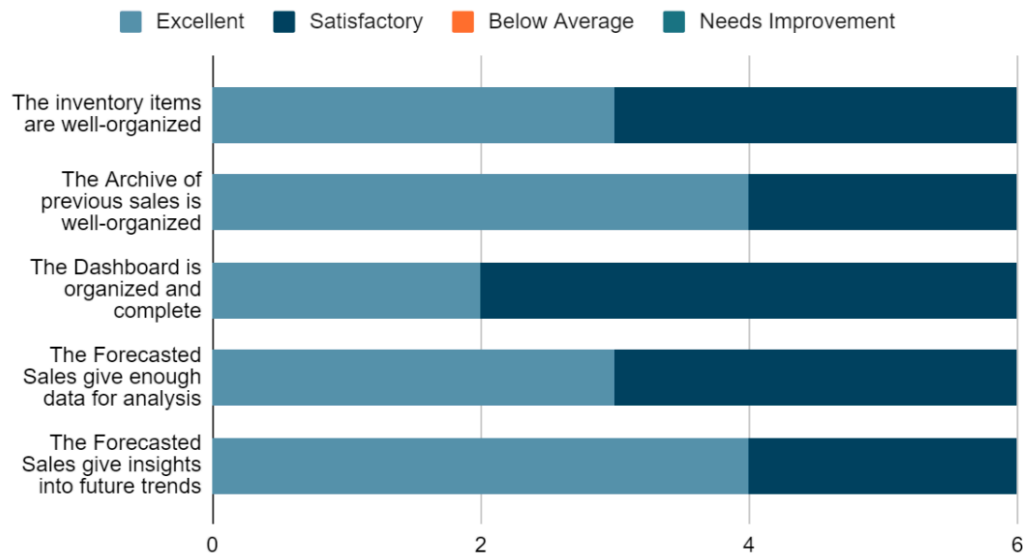


Figures 5.2.9 showed the respondents' evaluation of the Time behaviour. The respondents had an average of 3.944 for the entire category on the smart inventory system, which is an average interpretation of Excellent for the system's Time behaviour. They believed that the system was able to give results in a timely manner.

**Figure 5.2.10**

Graph of the Respondents evaluation on the Functional Completeness of the Smart Inventory System

Functional Completeness - Smart Inventory



Figures 5.2.10 showed the respondents' evaluation on the functional correctness of both systems. The respondents had an average of 3.5 for the entire category on the traditional system, which is an average interpretation of Excellent for the system's Functional Completeness. They believed that the system had the complete requirements that they needed.



Paired Samples T-Test

Paired Samples T-Test

			statistic	df	p
FUNCTIONAL CORRECTNESS (2)	FUNCTIONAL CORRECTNESS	Student's t	1.100	5.000	0.321
OPERABILITY (2)	OPERABILITY	Student's t	2.805	5.000	0.038
SECURITY (2)	SECURITY	Student's t	2.193	5.000	0.080
SCALABILITY (2)	SCALABILITY	Student's t	2.454	5.000	0.058

Note. $H_a: \mu_{\text{Measure 1}} - \mu_{\text{Measure 2}} \neq 0$

P < 0.05 indicates significant difference

Table 5.2.1

Table of Paired T-Test Samples

Table 5.2.1 showcases the T-Test calculation created by the statistician assigned to the group. Based on the legend of p for significant difference, there is only one significant difference between the two systems. The one category that had a significant difference across the two systems is operability with a score of 0.038. One possible factor on why the significant difference cannot be seen that much is possible from the low count of respondents.



CHAPTER VI

CONCLUSION AND RECOMMENDATION

6.1 CONCLUSION

The Smart Inventory System was created with the use of SARIMAX to properly predict bulk manufacturing and sales in the primary beneficiary Emdee Chemical Marketing Corporation. The research gap in the current literature and the reason why SARIMAX was used was because it was seldom used for bulk buying, which is why it was the algorithm used to predict the company's sales. The researchers limited the use of using five years of sales data for prediction to ensure accuracy and validity. The system was made to be a web application to be mainly accessed using a computer. Once it was deployed and tested by the beneficiary, they were able to note that the sales forecasting was able help with how they would market or produce items. One such way is by ramping up marketing during high and lows to further increase the sales while decreasing possible loss. The system being deployed online would allow the company to use it anywhere and was able to archive their invoices in the database. One limitation that they noted with the traditional system was that they needed to contact or go to the office to see the current inventory count. Based on the accuracy score and the trend line that the model was able to receive, the model was able to somehow predict the highs and lows during bulk buy. It was able to predict the trendline of the sales when comparing the prediction to the test data. During testing the



beneficiary noted that with the use of the system archiving invoices and keeping track of inventory was easier due to it being hosted in the cloud. With the use of the system the beneficiary was able to test how they would use in in their day-to-day operations. Especially during heavy or light workloads and workdays the system was able to handle the load excellently. The respondents also noted that the system can handle tracking the inventory during big orders in an excellent fashion. In terms of bulk sales, it was able to predict the trend of going up and down accurately, but the value of the prediction itself would not be accurate. Overall, the system was able to improve the user experience of the admins and employees, mainly in the operability category where there was a significant difference between the two systems. Based on the feedback from the beneficiary they believed that the system would be able to help them to be more on top of keeping track of the inventory. With the system giving proper notifications on low inventory, they were able to take note of these and create plans on how to solve this, which in turn helped with the slow turnover. Lastly, the beneficiaries confirmed that the smart inventory system would ensure 90 - 100% accuracy in inventory counts.

6.2 RECOMMENDATION

The proponents of the study suggest future studies to look into implementing logistics to the system, using Radio Frequency Identification (RFID) for barcode scanning of the products to enhance the accuracy of the inventory management,



speed-up business operations, and for real-time data communication and analysis with the system. The proponents also suggest integrating other departments from the company to the system as currently the system only deals with the industrial side, as there is so much more to explore with the cosmetics, accounting, and logistics of the company.

The beneficiary also recommended the following from the suggestion portion of the survey.

- Separate database for raw materials.
- For the invoices add another check box for the shipping fee and an additional column for the Value Added Tax (VAT).
- Add a module for purchasing and integrate the invoices to it
- Once purchasing has been integrated have the inventory module have a connection where once something was purchased and added to the invoice it would automatically subtract from the inventory.
- They also mentioned that we should add a section to differentiate the raw materials to the finish goods.

The beneficiary recommends the following to the proponents as seen in Appendix H.



- Implement Stock Keeping Units (SKU), wherein the system checks the monthly stocks of an item and keeps track how many were sold and how many were remaining and the remaining are carried over to then next month.
- Implement an owner or master type of account, that can oversee all the operations of the company and has an activity log screen, wherein the owner can see what invoices were made, the customers added, and what are the day-to-day operations of the company.
- Implement a messaging feature between admins and a way to notify or keep track of whenever an admin is logged and his or her activities.
- Add another database where all of the deleted entries go for either restoration or deletion.



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APPENDIX A – SIGNED QUESTIONNAIRE

Traditional Logbook Inventory System Evaluation Tool
Adopted from Camaclang, Estrada, Tielao & Rosas 2022 with modifications

NAME: (Optional) _____
AGE: (Optional) _____
GENDER: (Optional) _____
DEPARTMENT: _____

Direction: Please rate the following characteristics by putting a check on the column that best describes your observation of each of the following criteria.

SCALE:			
4 - Good	3 - Satisfactory	2 - Below Average	1 - Needs Improvement

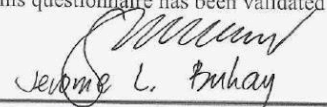
Software Criteria based on ISO 25010:2023	4	3	2	1
FUNCTIONAL CORRECTNESS				
1. The system can be used to archive old sales				
2. The system was able to accurately monitor future trends				
3. The system allowed me to keep track of my inventory				
SECURITY				
4. The system ensures that data can only be accessed by admins or those who have access				
5. The system is secure from data being stolen				
6. The system has adequate safeguards in place to prevent accidental or unauthorized data modifications				
SCALABILITY				
7. The system works well during days with heavy workloads				
8. The system works well during days with light workloads				
9. The system can handle tracking inventory during big orders				
OPERABILITY				
10. The system allows me to track my inventory easily				



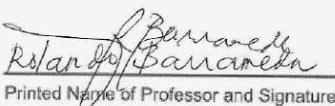
11. The system allows me to track my sales easily					
12. I can use the system without the need for assistance					

Recommendations/Comments:

This questionnaire has been validated by the following university faculty:


Jerome L. Puhay
 Printed Name of Professor and Signature

10-07-24
 Date Validated


Rostand B. Baraneta
 Printed Name of Professor and Signature

10-07-24
 Date Validated



*Smart Inventory System: Sales Analysis using SARIMAX for a
Chemical Manufacturing Company Evaluation Tool*
Adopted from Camaclang, Estrada, Ticlaio & Rosas 2022 with modifications

NAME: (Optional) _____
 AGE: (Optional) _____
 GENDER: (Optional) _____
 DEPARTMENT: _____

Direction: Please rate the following characteristics by putting a check on the column that best describes your observation of each of the following software criteria.

SCALE:			
4 - Good	3 - Satisfactory	2 - Below Average	1 - Needs Improvement

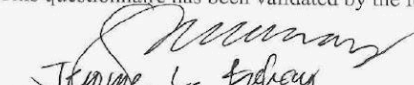
Software Criteria based on ISO 25010:2023	4	3	2	1
FUNCTIONAL CORRECTNESS				
1. The system allowed me to keep track of my inventory in real-time				
2. The system was able to accurately predict previous sales				
3. The system was able to accurately monitor future trends				
4. The system can be used to archive old sales				
OPERABILITY				
5. I can see the notifications easily/clearly				
6. I can use the system without the need for assistance				
7. Low Inventory notifications by the system are helpful				
8. I can easily see the inventory items clearly				
9. The system has necessary shortcuts				
FUNCTIONAL COMPLETENESS				
10. The inventory items are well-organized				
11. The Archive of previous sales is well-organized				



12. The Dashboard is organized and complete				
13. The Forecasted Sales give enough data for analysis				
14. The Forecasted Sales give insights into future trends				
SECURITY				
15. The system ensures that data can only be accessed by admins or those who have access				
16. The system is secure from data being stolen using Bcrypt hashing				
17. The system has adequate safeguards in place to prevent accidental or unauthorized data modifications				
SCALABILITY				
18. The system works well during days with heavy workloads				
19. The system works well during days with light workloads				
20. The system can handle tracking inventory during big orders				
TIME BEHAVIOUR				
21. The system predicts future trends fast				
22. I can access archived data fast				
23. The inventory system responds quickly to my inputs				

Recommendations/Comments:

This questionnaire has been validated by the following university faculty:


 Printed Name of Professor and Signature

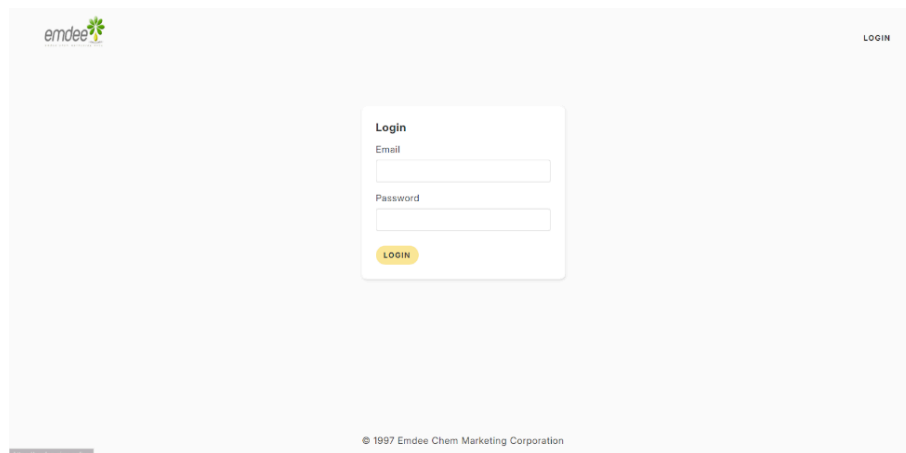
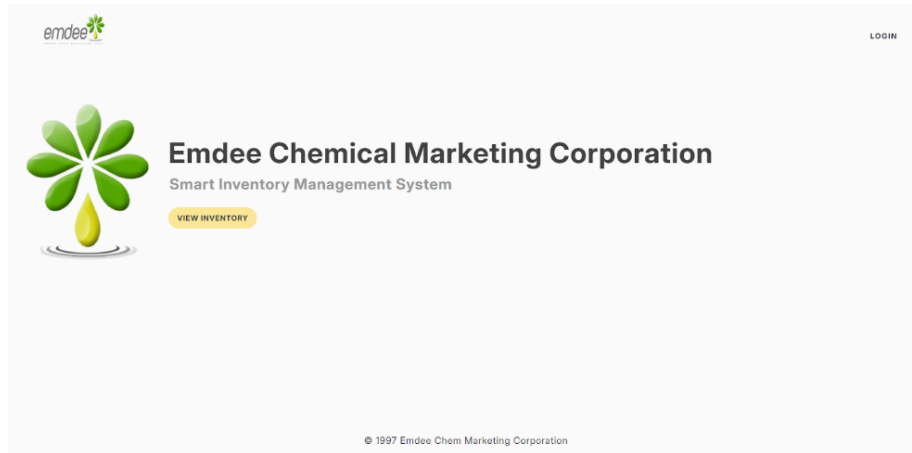

 Printed Name of Professor and Signature

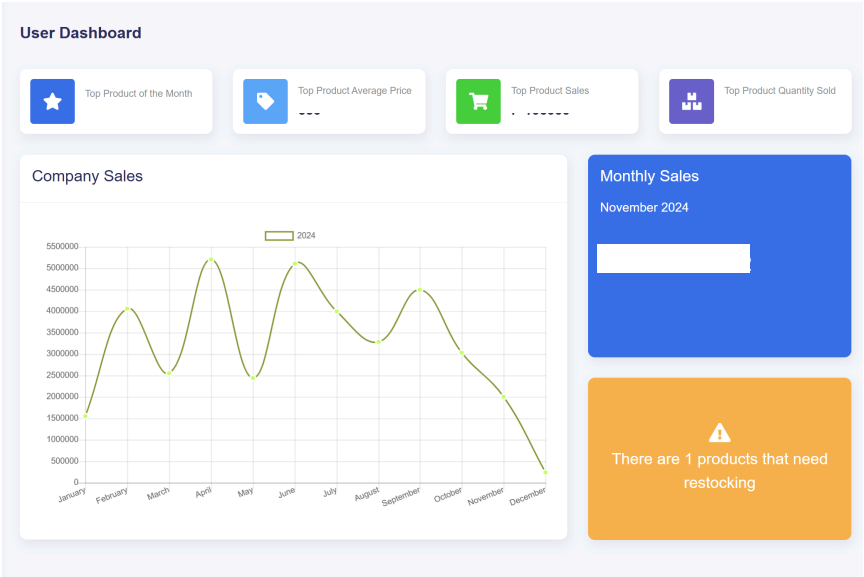
10-07-24
 Date Validated

16-07-24
 Date Validated



APPENDIX B - SCREENSHOTS OF THE APPLICATION





Emdee SIMS

Dashboard

Inventory

Customer

Product

User Customer

Show 10 entries

Search

CUSTOMER NAME	T1	CUSTOMER TYPE	T1
		Company	
		Company	
		Company	
		Company	
		Company	
		Individual	
		Company	
		Company	
		Company	
		Company	

Customer Name

Showing 1 to 10 of 156 entries

Previous 1 2 3 4 5 16 Next

Encarnacion Nedran Vilanueva

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For the Completion of an Undergrad Thesis

Recent Invoices

INVOICE FILE	INVOICE DATE	CUSTOMER NAME	TOTAL PRICE
	2024-05-03		
	2024-04-30		
	2024-04-30		
	2024-04-30		
	2024-04-29		
	2024-04-26		
	2024-04-26		

Encarnacion Nedran Vilanueva

© 1997 Emdee Chemical Marketing Corporation

For the Completion of an Undergrad Thesis

User Accounts Table

Add Account

Show 10 entries

Search:

First Name	T1	Middle Name	T1	Last Name	T1	Suffix	T1	Mobile Number	T1	Email	T1	Action
CEDRIC		C		NEDTRAN								✎ ✖
CHARLES		D		VILLANUEVA								✎ ✖
JULIAN		A		ENCARNACION						julianencarnacion19slm@gmail.com		✎ ✖
MARIA		D.		DOE						user@gmail.com		✎ ✖

Showing 1 to 4 of 4 entries

Previous

1

Next

 Deleted! Account has been successfully deleted. OK	 Success Invoice updated successfully! OK	 Success Customer created successfully! OK	 Success The entry has been successfully edited! OK
 Success Invoice created successfully! OK	 Success The entry has been successfully edited! OK	 Success Account created successfully! OK	 Are you sure? You won't be able to undo this! Yes, delete it! Cancel
 Deleted! The customer has been successfully deleted. OK	 Error Cannot delete customer because it is referenced by 1 invoices OK	 Success Product created successfully! OK	 Deleted! The product has been successfully deleted. OK



Create Invoice

Invoice Details

Customer Name

INVOICE DATE

yyyy-mm-dd

View Invoices

Follow the first row format click the button on the right add a new row

Re-Order

Add Product

PRODUCT NAME

Select Product Name

PRODUCT TYPE

Select Product Type

UNIT PRICE

Input Unit Price

QTY

0

SIZE

Select Size

TOTAL PRICE

Calculated Total

PRODUCT NAME

Select Product Name

PRODUCT TYPE

Select Product Type

UNIT PRICE

Input Unit Price

QTY

0

SIZE

Select Size

TOTAL PRICE

Calculated Total

Discount Percent

%

Discount: ₱0.00

Sub Total Price: ₱0.00

Create Invoice

Final Price: ₱0.00

Emdee SIMS

Dashboard

Admin

Create Account

Accounts

Create Invoice

Inventory

Customer

Product

Invoice

Machine Learning

Sales Forecasting

Admin Invoice

Show 10 entries

Search

FILE NAME	INVOICE DATE	CUSTOMER NAME	DISCOUNT	TOTAL PRICE	DOWNLOAD	ACTION
appendix1	2024-12-10	JANE DOE	10	90		
	2024-05-03		0			
	2024-04-30		0			
	2024-04-30		0			
	2024-04-30		0			
	2024-04-29		0			
	2024-04-26		0			
	2024-04-26		0			
	2024-04-26		0			

Edit Invoice Entry

FILE NAME

appendix1

INVOICE DATE

12/10/2024

CUSTOMER NAME

JANE DOE

Reselect the Product Types

Add Product

PRODUCT NAME

PRODUCT TYPE

UNIT PRICE

1

QTY

100

SIZE

cby

TOTAL PRICE

100.00

Discount

☒

10

%

Discount: ₱10.00

Sub Total Price: ₱100.00

Total Price: ₱90.00

Save changes

Close



Emdee SIMS

Dashboard

ADMIN

- Create Account
- Accounts
- Create Invoice

INVENTORY

- Customer
- Product
- Invoice

MANAGE LEARNING

- Sales Forecasting

Admin Customer

Show 10 entries

Search

CUSTOMER NAME	CUSTOMER TYPE	EMAIL	ADDRESS	ACTION
AAENCAR	Individual	juliancamaron199n@gmail.com	Taguig	Edit Delete
	Company	no email	Quezon City	Edit Delete
	Company	qwert@mail.com	Makati	Edit Delete
	Company			Edit Delete
	Company			Edit Delete
	Company			Edit Delete
	Individual			Edit Delete
	Company			Edit Delete
	Company			Edit Delete
	Company			Edit Delete

CUSTOMER NAME CUSTOMER TYPE ADDRESS EMAIL ACTION

New Customer Entry

Create a new customer entry by filling all the necessary fields.

CUSTOMER NAME

Input Customer Here

CUSTOMER TYPE

Select Customer Type

EMAIL

Input Email Here

ADDRESS

Input Address Here

Add

Close

New Product Entry

Create a new product entry by filling all the necessary fields.

PRODUCT NAME

Input Product Name Here

PRODUCT TYPE

Input Product Type Here

STOCK

Input Stock Here

STOCK THRESHOLD

Input Stock Threshold Here

SIZES

SELECT SIZES HERE

Add

Close



Admin Product

Show 10 entries

Search:

PRODUCT NAME	PRODUCT TYPE	STOCK	THRESHOLD	SIZE	LAST BOUGHT	ACTION
		99	1	liter		✎ ✖
		59	0	cty		✎ ✖
		78	3	pail		✎ ✖
		140	3	bottle		✎ ✖
		100	84	100 ml, bottle, gallon, piece		✎ ✖
		124	0	liter, gallon		✎ ✖
		72	0	liter, gallon		✎ ✖
		81	24	gallon, pail		✎ ✖
		104	0	drum		✎ ✖
		115	0	liter, gallon		✎ ✖

PRODUCT NAME PRODUCT TYPE STOCK THRESHOLD SIZE LAST BOUGHT ACTION

Edit Product

PRODUCT NAME

AAAACHEM

PRODUCT TYPE

STOCK

99

STOCK THRESHOLD

1

SIZES

SELECT SIZES HERE

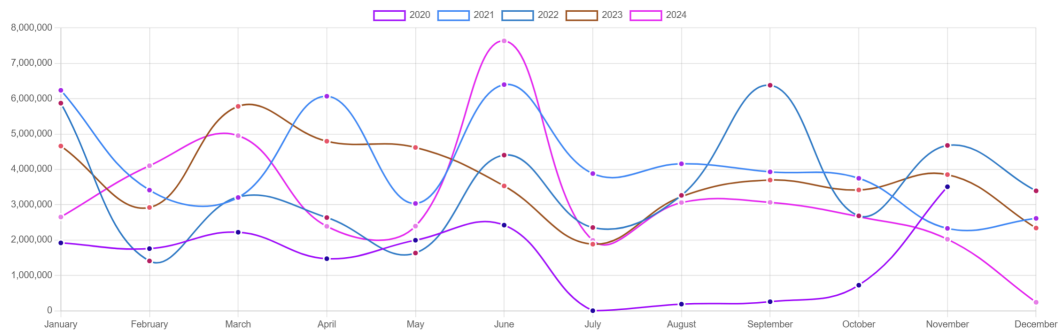
Save changes

Close



Sales Forecasting

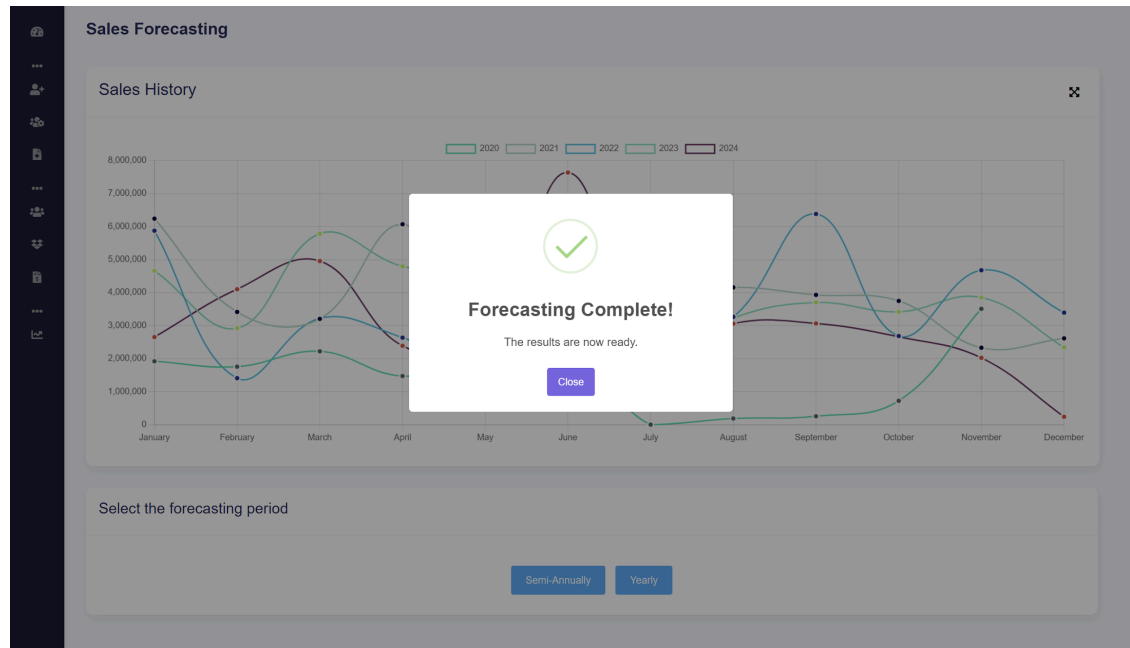
Sales History



Select the forecasting period

Semi-Annually

Yearly



Historical Accuracy

Mean Absolute Error (MAE): 33323.09

Root Mean Squared Error (RMSE): 82769.91

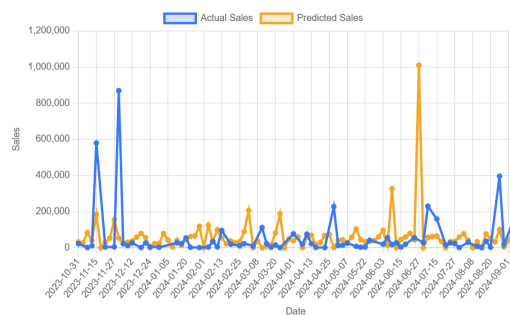
Scaled MAE: 0.27

Scaled RMSE: 0.33

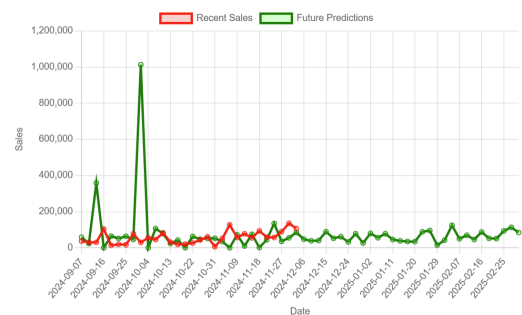
Mean Arctangent Absolute Percentage Error (MAAPE): 0.5447838880505516

Scaled MAAPE: 0.20781647942463002

Historical Sales



Forecasted Sales





Future Accuracy

Mean Absolute Error (MAE): 81828.4

Root Mean Squared Error (RMSE): 195917.64

Scaled MAE: 3.6

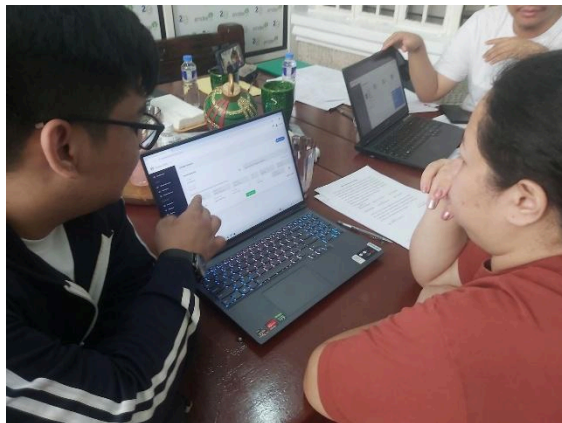
Scaled RMSE: 4.52

Mean Arctangent Absolute Percentage Error (MAAPE): 0.6899610044831072

Scaled MAAPE: 0.5557555633597807

Forecasted Sales in the next 180 days

DATE	FORECASTED SALES	FIRST DIFFERENCE	FORECASTED SALES GROWTH
09/07/2024	57741.54	-29025.00	-27.38
09/10/2024	23646.08	-34095.46	-59.05
09/13/2024	358455.27	334809.19	1415.92
09/16/2024	0.00	-358455.27	-100.00
09/19/2024	64137.14	64137.14	64137138850000000.00
09/22/2024	51108.48	-13028.66	-20.31
09/25/2024	63537.17	12428.69	24.32
09/28/2024	46021.95	-17515.22	-27.57

**APPENDIX C - EVALUATION PHOTOS**

**APPENDIX D - ETHICS CERTIFICATE**



De La Salle University - Dasmariñas
DLSU-D RESEARCH ETHICS COMMITTEE

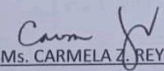


CERTIFICATION

The DLSU-D Research Ethics Committee has reviewed the protocol for the study titled **"SMART INVENTORY SYSTEM: SALES ANALYSIS USING SARIMAX FOR A CHEMICAL MANUFACTURING COMPANY"** code **DERC_24-25_061T** as proposed by **JULIAN KRISTOFF ENCARNACION, CEDRIC JOHN NEDTRAN and ALVIN CHARLES VILLANUEVA**, College of Information and Computer Studies.

The committee approves the implementation of the study. This certification is valid until April 30, 2025.

Issued this 4th of November 2024.


Ms. CARMELA Z. REYES
DLSU-DERC Chair



APPENDIX E - STATISTICIAN CERTIFICATE



De La Salle University
D A S M A R I Ñ A S

College of Science
Mathematics and Statistics Department
(+6346) 4164524 / (+632) 8447832 local 3087
www.dlsud.edu.ph



December 12, 2024

Date

CERTIFICATION

The DLSU-D Data Analysis and Training Center has reviewed and validated the
questionnaire and results of the study titled
Smart Inventory Management System: Sales Analysis Using

SARIMAX for a Chemical Manufacturing Company

Research Title

as proposed by **Julian Kristoff A. Encarnacion** **Cedric John C. Nedtran**

Name of the Researcher/s

Alvin Charles D. Villanueva

College of Information and Computer

Studies. The statistician and the unit head of DATC approves the implementation of the
study.


Mr. JEROME L. BUHAY

Unit Head, DATC DLSU-D – Statistician

Nurturing innovation and nature.



APPENDIX F - PLAGIARISM SCAN (TURNITIN)

ENCARNACION-NEDTRAN-VILLANUEVA-CHAPTERS 1-6 MANUSCRIPT

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De La Salle University - Dasmariñas

**SMART INVENTORY SYSTEM:
SALES ANALYSIS USING
SARIMAX FOR A CHEMICAL
MANUFACTURING COMPANY**

Submitted in Partial **Fulfillment**
Of the Requirement for the Degree
Bachelor of Science in Computer Science
With Intelligent Systems Track

Thesis Adviser
Isabelinda D. Manzano

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A	B	C	D	E	F	G	H
JANE DOE						2024-12-11	
10 liter	ALCOHOL				50.00	500.00	
5 pack	DISPOSABLE FACE MASK				20.00	100.00	
3 bottle	CAUSTIC SODA				270.00	810.00	
NOTHING FOLLOWS							
						Discount	10%
						Discounted Price	141.00
						Original Price	1410.00
						Total Price	1269.00



APPENDIX H - TRANSCRIPTIONS OF THE INTERVIEWS WITH THE COMPANY

April 30 - Emdee Corporation Interview Transcription

Legend

- JE* - Julian Kristoff Encarnacion
CN - Cedric John Nedtran
AV - Alvin Charles Villanueva
V - Sales Representative Admin
R - Quality Assurance Admin

JE - Hello and Good morning po. Ma'am and Sir we are students from La Salle Dasmariñas and uh we are here to interview you po for the purpose of our thesis, to know more about the company and know more about uh yung mga kailangan po ng system. So uh first po ang system po namin or ang paper po is uh about smart logistics system po uh which may automated na po siya na pwede na ninyo po doon ilagay ang items niyo na nakaparang advanced version po siya ng excel yung gusto naming gawin.

V - Okay

JE - Na may pagkaautomated na po siya, so plano po namin na meron po siyang mga inventory alerts po na kung kunyari paubos na po ang stocks niya pwede niya po magaalert po siya. Additional nalang po is meron po kaming idadag na sales analytics so kapag kunyari po napansin ng system is yung peak po na months is for example mga June or July magnonotify naman na po siya and ang mga peaks naman po.

CN - Wait lang okay lang ba if I may add?

JE - Sige

CN - If I may add po sa naisip po naming system po kasi syempre hindi lang po yung parang um.. ah... inventory management system na marami na pong, na marami na pong existing gusto po namin lagyan po ng twist po. So sa sinabi po ni Encar na may smart logistics at yung sales forecasting ang balak



rin po namin implement po na makakatulong po sa inventory, makakatulong rin po sa pagmanage ng po warehouse in the sense na pagimplement po ng Internet of Things. Ah..a.. Alam po ninyo yun yung IoT?

V - No, not yet

CN - So to explain ang IoT imagine po yung mga devices po interconnected sa iisang WiFi, so for example sa WiFi ng Phone connected sa WiFi ng aircon, connected sa WiFi ng Electric Fan, so with that interconnected kunyari sa phone pwede mo na i-off ang electric fan at i-off ang aircon and all that.

V - Okay, interconnected ano tawag dun?

CN - Internet of Things

JE - Or IoT po so... bale plano po namin siya ma-implement din sa system po para sa company is ang mga RFID po so kunyari po para instead na manual counting may RFID nalang din po para mas automatic nalang po siya mag update sa system. Kasi

V - That's nice

JE - Opo, natanong ko na rin kay tito, and na naverify na po niya na is wala pa po kayong system in place na so parang manual na ata.

V - Excel

JE - Excel lang po, so ayun po with that po

V - Nag automated pa

JE - With that po mas magiging automated naman na po siya and gusto po namin

V - Magiging helpful kayo saamin.

JE - Opo ma'am

V - *laughs*

JE - So yun din po ang pinaka goal namin is to help nalang din po so if matanong nalang din po namin okay lang po ba maintroduce niyo po yung company? Or pa ano nalang po pa brief background nalang din po?



CN - Ay before that po okay lang ba po pwede ba po namin makuha namin ang permission dito po sa thesis namin para po nasa record na po gamitin ang pangalan ninyo and yung company kasi po ganun po optional lang naman po syempre we... ang data po ninyo, you're free to do whatever you want with that.

JE - Bale ano nalang din po kasi po ma'am um nagtatanong po kami ng permission is pwede ba po namin masabi yung pangalan ng company.

V - Ang pangalan ng company? Sige i-balikan kita diyan i think kung sinabi ni Sir Calo na okay gamitin ang Emdee Chem wala namang problema

JE - Kasi okay lang naman din po kung kahit hindi

V - Pero ang confirmation kay Sir Carlo ko nalang i-coconfirm na okay lang gamitin sa Thesis ninyo ang Emdee Chem.

JE - Sige po uh... with that po moving forward. Pwede pa brief background lang po ng company ninyo.

V - Ah, brief background. Si Emdee Chem started at uh... registered siya August 1997, so twenty-six years na this year. So catering mga industrial side ng mga tawag dito... yung mga oils, mga greases, yung po yung ang mga uh... ano ang mga product ng Emdee. So ngayon kasi nagkaroon na ng mga skincare kung ano mga twenty seven years na siya catering mga multi national companies na tulad na sabi ni Miss Grace.

JE - So uh... with that po ano po yung masasabi niyong mga challenges na na-face ninyo po as a company, specifically po doon sa logistics side nalang po.

V - Sa logistics side, hindi kami actually nagkaroon ng problem since yung mga ano uh... client natin is within ano lang naman Laguna, Cavite mga ganyan and doon naman galing ang warehouse namin. So logistically parang wala naman siyang ganung problema. Sa documentation siguro sa paghahandle ng mga ano

JE - Papel?

V - Sa... sa system yun yung challenge pero sa in terms ng pagdedeliver wala kaming, mga zero delay kami, awarded kami ng mga multi-national



companies na zero delay kami. Kasi yun naman ang goal talaga, kailangan mabilis. So yun yung ano ni Sir Carlo kasi ang pinaka theme niya na zero delay dapat si Emdee. So yun.

JE - Pagdating po dun sa pagstorage po ng mga items, ano po yung mga problems po na na-face ninyo siguro or yung pagkeep track po ng mga items po

V - Sa archiving, kasi uh... ang warehouse na naman namin kasi standard naman siya na nakafollow din sa kung paano ang mga flow ng mga international na ano, yung standard na siya kaya lang yung sa... tracking lang paghalimbawa kailangan ninyo three years backward. So wala kaming data kasi manual ang archive namin. So kailangan mo talaga kalkalin yung paper. Wala siyang data na naka nasa system. So ayun walang soft.

JE - Sa ganun po ba naaffect po ba yung decisions niyo as a company kasi hindi ninyo nakikita ang data beforehand.

V - Hm... Siguro minimal lang naman kasi ang babalikan mo lang naman doon si yung mga pricing, mga kung ano, uh... gaano ba ka recent itong order kasi may like client na nagorder lang this year and then following year na naman yung mga ganon.

JE - Pero ano po ma'am uh, definitely important din naman

V - Important talaga siya lalo na sa panahon ngayon. *laughs*, kaya looking foward din kami maadapt namin yung ganung sistema para kami rin dumadami ang mga workload, so nageexpand na rin kasi si Emdee, may iba na siyang industry. Dati kasi si ano lang yung sa mga chemicals lang kami kasi ngayon na may skincare na nasa natural ano na rin kailangan na rin naman mag adopt sa bagong sistema. Which is very helpful kung automated na siya.

JE - And uh... tanong nalang din naman natin yung Charles about sa Machine Learning yung mga ganon tanong ba natin?

AV - Sige sige

JE - So ayun tatanong lang din namin ma'am kasi parte po siya ng paper namin is aware ba kayo sa machine learning? Yung mga ganon po.



V - Hindi pa masyado no, parang wala pa. Enlighten us with that.

JE - Sige po. So to brief you nalang po, parang AI isipin ninyo mga AI.

V - Mga AI oo.

JE - Parang AI siya na lets say na nagkinakain po yung model yung machine learning model ng data na three years back na sales data. So pwede po naming gawin is gagawa po kami ng system na gamit ang three years back na data na makakaforecast po siya ng possible future sales.

V - Hmm...

JE - So na experiment naman na po namin dati ang so far okay naman po ang accuracies niya, so kahit hindi naman po siya super accurate magagamit naman po siya as ng company na moving forward po alam po yung mga possible na mangyari. So it can help nalang din po making with decision making doon sa business side.

And then uh... ano may mga questions paba kayo with specifically ano yung mga need sa system natin?

AV - Ayun ang mga features if may hinahanap po sila.

JE - Isa rin pala yun if may hinahanap or may mga features ba kayong gustong masama or hinahanap sa isang inventory system?

V - Ang pinakaneed ko talaga is yung inventory system, oo yun ano yun parang crucial yun kailangan meron talaga nun. Tas yun sa mga incoming, outgoing na ano kung anong system yung tawag dun. Pero ngayon kasi manual at tsaka kailangan din namin ng manual kasi ang sa audit ang tinitignan nila is yung sa paper, hindi yun sa ano. So ang incoming namin naka log book ang outgoing namin naka log book rin. Kung yan din kasama sa ma-systemize that would be better.

JE - Sige po so when you say the incoming and outgoing, incoming po yung mga...

V - Deliveries, raw materials, packaging

JE - Ano po usually ang mga incoming po ninyo?

V - Yun mga raw materials and packaging yun ang mga incoming namin.



JE - Sige po and then uh... out going po ang mga...

V - Outgoing po ang mga finished, yes ang mga finished products

JE - Okay so kung madadagdag din po namin, is ano po ang mga products na... siguro nahihirapan po kayo, or matagal po ang shelf life lets say, or wala naman po?

V - Wala naman two years lang naman, ang skincare two years lang ang shelf life niya eh so.

JE - Pero may mga cases na po ba na tumagal na po siya doon sa warehouse, na hindi pa nasesell

V - Oo kasi meron mga hindi napansin, since wala nga siyang systematic inventory so may mga naligtaan.

JE - May mga cases naba po kayo na doon nagkaroon ng loss na po?

V - Oo may loss na doon definitely, like nageexpire yung raw mats mga loss yun.

JE - So with that po if ever we can dagdag a system na magsasabi ng kailang yung huling purchase or yung expiring. Dagdag lang po namin definitely.

So so far papakita ko lang din ang nagawa kong prototype.

V - Okay sige excited *laughs*

JE - Pero medyo basic pa siya wala pa ang ibang details po

V - Siguro pag ka graduate ninyo ano na yan o ngayong year niyo ba yan i-lalaunch

JE - Baka mga next year na namin

V - So kami ang unang users?

JE - Opo ang gusto po namin ma push through kasi malaking tulong po saamin.

Other than that may mga ibang questions ba po kayo saamin? Specifically para sa SOP natin? Yung mga gusto natin madagdag sa SOP?



CN - Check ko lang ang kung ano pa ang gusto natin, pero noong nagpaconsult po kami sa adviser, ang sinabi niya po. Yung statement of the problem hindi ito nareresearch mainly kung ano lang po ang problema ninyo eh. So sabi rin po ninyo ngayon na kailangan ng inventory system at sabi po kailangan ninyo mag adapt to the times diba kasi maiiwanan kayo if you stay sa current na system.

V - Tama, sales data mga ganun, archiving

AV - Ilan usually ineexpect na user na...

JE - Isa rin pala yun, if ever po nagawa na po namin sino po ang usually gagamit.

V - Ako, uh... mainly sa office yun eh, sa warehouse, sa office. So mga key person yan eh mga at least 5. Mga warehouse, logistics, admin, tapos syempre sa mga boss kasi may approval like mga...

JE - Sige po so if ever ang mga ganun, um... magkakaroon po ng mga levels if ever ang mga admin tapos mga may user lang po. So if ever po may admin kayo po ni sir?

Okay po so isip namin sa ganoong admin level kayo po ang pwede nalang mag delete ng mga...

V - Yes

JE - Pero sa mga pagedit po siguro automatic nalang or pwede din yung mga ano... users?

V - Dapat ang mga archiving you're like mga 3-5 years data, Oo ganun.

JE - Sige po kasi naisip nalang din po namin kung ano ang mga kailangan po na, kunyari sino po ang gagamit uh... bibigyan lang namin sila ng mga anong pwedeng nilang gawain

V - Yes, opo hindi rin pwede lahat yun kasi mamaya baka may maburang importante.

JE - Opo mga ganun po and definitely may mga sensitive na information naman po and moving on po sa sensitive information po ang we are also here to ask po for the sales data and inventory data nalang din po. So ang sales data



na yun din po uh... kung hanggang saan po kaya kasi sabi po ninyo is mahirap mag archive yun din po.

V - So mabibigyan kita ngayon ng this year and last year siguro na uh... ano sabi ko nga na manual yun so kayo

JE - Okay lang po, and gamit po noon ng manual po uh... okay lang po ba i-type po namin and encode namin.

V - Type niyo nalang

JE - Hindi po namin isasama ang parang...

V - Company name?

JE - Yes po. Yung kailangan lang po namin is kailan lang po siya, yung sales, yung date, yun lang po kasi yun lang naman ang kailangan doon sa step. Tapos so no worries naman po sa more confidential side.

V - Oo sa marketing naman yun, so yun lang ang naremind saamin na may ganung mga info.

JE - Kailangan paba po namin mag sign parin? Or...

V - Hindi na, okay na kasi sabi naman ni Sir Carlo na okay na.

JE - Okay sige pero sa side nalang din po namin hindi na naman namin ididisclose po, definitely lalagay nalang po namin sa paper na hindi namin ididisclose. I-stastate nalang po namin for the record is we will only use date, yung sales, siguro pati ang price sama nalang din namin.

CN - Tawag dito, since kung susulatin natin ang product name gamit nalang tayo ng mga Product A, Product B, and Product C...

V - Ah... oo mga...

JE - Most likely mga Product A and Product B and then additionally nalang din po would probably be the yung... amount ng inventory at that time, so para lang may dagdag na data kami. Or kung hindi po kahit yung inventory nalang.

V - Yung inventory madugo yun *laughs*

JE - Madugo po ba talaga?



V - Madugo talaga yun kasi parang ngayon ko palang siya cino-collate yung inventory sa kasi napaka dami niya like ang dami niyang departments so sila ni raw materials iba yun. Ibang inventory yun, mga finished iba yun. Ibang inventory yun. So mga packaging ibang inventory yun. So ang dami niyang iba ibang. So pag yung sistem ninyo in place mas madali nalang saakin yun *laughs*.

JE - Sige po if ever...

V - Malaking tulong ang gagawin ninyo talaga actually

JE - Sige po if ever po ganun po uh... gagawa nalang po kami ng seperate na dashboard per department, if possible po. So kunyari uh... isa po for marketing siguro sa kanila more on the sales...

V - Sales aspect, tama

JE - So kapag doon naman doon po sa raw materials, sila po may kopya sila na yung inventory na puro raw materials lang naman. Hindi na kasama ang finished products.

V - Yes magkakahiwalay dapat

JE - Tapos sa side nalang din po ng admin, sanyong dalawa po is makikita naman po ninyo lahat.

So may additional question pa kayo na pwede natin madagdag?

CN - Ako pre may madadagdag sa dashboard, So sino po ang mga gagamit ng mga different dashboards po para at least na specify na. So may mga raw materials, marketing...

JE - So parang mga departments

CN - Ilan po ang mga total departments na gagawan po ng dashboard? If ever po?

V - Siguro mga 5. Warehousing, logistics kasi sila logistics maaapsahan nila ang mga finished na for delivery na. So logistics, warehouse, admin, and then sa QC ba kailangan ngayon?

R - Kailangan

V - Oo sa ano, kasi yung mga raw materials sila yun eh.



CN - Okay so ang raw materials from QC okay.

JE - So pwede matanong nalang din po namin uh... na additionally po is papaano po ang mga warehouses ninyo kasi sabi din po ni sir na dito mostly raw materials tama po ba?

V - Um hmm *nods*

JE - So papaano naman po ang mga ibang warehouses ano po ang mga relationship nila tulad dito po sa Cavite may warehouse kayo specifically for this material, so papaano po ang iyong sistema for that?

V - Ang communication talaga namin is through phone, so uh... may log book lang kami doon kung ano ang mga nakalagay so may slip din sila doon na pag may kinuha doon with ano yun yung... nilabas nila so nakalog din doon. So ganoon naka paper manual.

JE - If ever po emails possible po ba yun? Or mahirap po sa ibang warehouses dahil sa internet....

V - Sa internet tsaka wala pang admin doon, wala pang admin doon na naka-assign kasi warehouse lang talaga siya so may custodian lang doon na magrelease nung mga items na kailangan or mga raw mats na kailangan so yun.

JE - Out of all of your warehouses po may mga warehouses naman po kayo na may internet connection?

V - *Agrees*

JE - Meron naman po?

V - Actually may internet naman din sa Cavite yun nga lang syempre hindi reliable pagmay ka puro digital era doon *laughs*. Kung бага kailangan din i-manual at both kailangan doon

JE - So siguro po if ever isip lang din po namin is um... mag message po kayo kunyari sa Cavite na need ninyo ang product na ito.

V - *Nods*

JE - So siguro pagnasulat na po yung i-slip pwede po siya picturan tapos i-send sa admin and then si admin na po magiinput ng changes po

V - Pwede din, pwede din nun, tapos ma-uup nalang din pala sa kanila.



JE - Opo para ma confirm nalang din po at nasa system.

V - Yes pwede, pwede.

JE - Sige po and uh,,, additionally nalang po kung pwede naman matanong ilan po yung warehouses niyo and ano po yung kunyari mga warehouses na ito and ano ang mga materials specifically.

V - Ngayon may dalawa kami uh... magkahiwalay kasi, isa nasa Alfonso mga ano yun more on sa mga chemicals oo tapos meron rin dito sa QC sa kabila more on sa mga skincare naman.

R - Sa cosmetics.

JE - Okay po uh... sa raw materials naman po dito din ba po siya or meron din...

V - Depende sa raw materials, kasi kung raw materials for chemical dadalhin yun sa Alfonso, dadalhin yun sa Cavite.

JE - So kung ano po yung need

V - Yes kung ano ang need, kung ang pero ang *clicks tounge* delivery point ni supplier dito lagi. So ang supplier magdedeliver ng raw material kung eto ay para doon sa chemical na ano dadalhin sa Alfonso, tapos kung eto naman ay cosmetics dito lang pero ang delivery point dito lagi. Oo.

JE - So ang pinaka main niyo talaga

V - Oo dito sa head office.

JE - Okay po so take note lang din namin. So kasi if ever po uh... since sabi din po ninyo may limited connectivity since Cavite sa Alfonso po.

V - Taga Cavite kayo right?

JE - Opo eto po *points at Cedric* taga Imus. Ako po dati dito lang

V - Kamukha mo nga si Miss Jan eh *laughs*

JE - *Laughs* yes po ninang ko po yun. So ayun po, ano paba? Since ganun nga po ang system parang mas hindi reliable and WiFi if ever po doon. If ever po papatest namin sainyo dito po muna.



- V - *Nods*
- JE - Okay lang naman po ba yun?
- V - Yes dito lang
- JE - Sige po and uh... moving forward doon po sa mga items na ilalagay namin, okay lang po ba kayo na ilang items gusto natin mga 10?
- AV - Yeah, mga ilang... yeah...
- JE - Mga 10 or kasi yung gagamitin po namin or sa....
- V - Sa paper?
- JE - Sa paper po, opo
- V - 10. 5 non-moving and 5 moving.
- JE - So yun lang naman din po, may mga questions nalang din ba kayo?
- AV - Wait lang, ano po usually ang laman po ng mabibigay na data po saamin? Yung mga products po
- V - More on sa industrial tayo kasi yung cosmetics kakalaunch niya lang mga last year, so wala pa akong mabibigay na data. Hmmhm... More on sa ano tayo sa mga chemicals ang mga aano ninyo.
- JE - May questions paba kayo? CJ?
- CN - May do-double check lang ako.
- JE - Double check nalang ninyo bago natin i-end yung recording.
- CN - Siguro last thing nalang po kung ano ang name and title po ninyo. Kayo po si Sir Rowel at si Miss Yien
- J - Ako po ang pinaka general admin at sales representative at si Sir Rowel
- CN - Ano po sir ang position po ninyo
- J - Admin and QA siya.
- CN - Ah... quality assurance



J - Mhmm

JE - So most likely po kayo ang mga makakausap po namin

J - Yes kami ang makakausap ninyo, kami ang magbabatuhan ng mga documents. Kaya nandito siya kasi kailangan naman din talaga ng mga ano.

JE - Sige po ma'am malapit din naman po ako dito po so if ever po moving forward mas makikita naman din po tayo.

J - *Nods*

JE - Sa may congressional nalang din po kami

J - Ah nice, oo sige

CN - Ah sige kung wala nang further questions okay na?

JE - Yes okay na

AV - Oo

CN - End na natin ito at 9:44 A.M

December 30 - System Demonstration and Evaluation

LEGEND

JE - Julian Kristoff Encarnacion

CN - Cedric John Nedtran

AV - Alvin Charles Villanueva

V - Sales Representative Admin

JE - Hello eto na po ang official test natin with Sir !@#\$ and eto po yung front-end system, so eto po yung homepage, hindi pa po naka log-in, dito and yung inventory. So test out muna natin ah sir pwede po kayo po mag press dito. Either view inventory or log-in po.

Ayun since hindi pa po kayo naka log-in pag pinress po niyo ang view inventory, mapoprompt po kayo mag log-in, so for now gamitin muna natin si admin muna gamitin or si user?



CN - For the sake ng pacing, maganda user tapos...

JE - So ano ang mga details ng user dito?

CN - user@gmail.com

JE - So maiiba naman po ang email pero for now user muna gamitin namin.

CN - test12345

JE - Ayun po so mag load lang po siya, oh wait.

CN - Hindi ganyan lang talaga nagloload yan,

JE - Ah okay, and then refresh nalang?

CN - Oo refresh refresh lang.

JE - So eto po yung dashboard natin sa pinaka-simula makikita po natin dito yung top product of the month, uh... top product price, top product sales, and yung top product na quantity. So for now po, since yung huling data po namin nakuha is from May pa, pinaka-simula ng May makikita nalang po natin dito sa graph yung company sales is mababa lang po siya for the month of May, kasi po kakasimula lang po technically. And then monthly sales for the month of May is ganito po ang value niya. Bale makikita nalang po natin kung ano yung sales, so pwede lang po kayo maghover lang po. Pwede niyo rin po i-test lang po siya. So yun nag hover po kayo makikita po niyo yung values po.

R - Maganda ito kasi may historical data no?

JE - Opo, mapapakita po namin siya, specifically for the year po.

R - Oo.

JE - So since dashboard lang naman po siya. Parang super barebones lang po siya talaga, summarized, hindi po siya yung super ano. And since this is for user yung makikita lang po talaga nila, as yung napagusapan po noong huli, is kung yung mga basic po na data. So kung ano po yung top product, ano yung sales natin so far, monthly sales, and ano yung top product ng huling month so eto po yung mga top products natin. And then pwede din natin makita sa left side sa screen, kung sino-sino ang mga customers natin so makikita po natin



ang mga listahan ng customers. So pindot lang din po kayo if you want to explore.

R - Anong nakapaloob sa kanila?

JE - Alin po?

R - Ah hindi siya pwede i-double click?

JE - Ah hindi po, parang po siyang list lang po talaga,

R - Oh okay list lang siya talaga

CN - I guess do keep in mind po na editable po siya if you want to see kunyari per page ilang rows sa taas. Yan po kung gusto po ninyo ma-edit yun or kung may hinahanap po kayo na specific na customer use the search bar.

R - Ah okay yung search dito. Gets gets.

JE - Opo and that's basically for yung customer, now lets go to product. So for the product eto lang po yung mga simpleng product table lang po kung ano yung mga products natin, alphabetical and ano yung type niya, stock, and description. So for the product type naman po, since yung huli naman is wala naman po kaming nakuhang masyadong product type sainyo, na leave nalang po namin blank and same nalang din po siya sa description since usually eto po yung nangagaling po sainyo.

R - Opo

JE - So for now yun nalang din po nun. So ganoon nalang din po sa stock, kayo nalang din po makikipaglagay ng laman po na yan pero may mga products po na may food-grade na may lubricant, ganyan lang po siya.

R - Pero paano mo siya maeedit na ano na ilan yung nakaschedule na?

JE - Ayun po, so for the edits naman po, since user lang naman po siya.

R - Ah hindi pwede

JE - Hindi po siya pwede since si Admin lang po.

CN - Si admin lang pwede



JE - Kaya po may dalawa po tayo, para ma-secure nalang din po ang system. Kasi hindi po natin alam kung user lang, may magalaw bigla.

R - Okay, gets gets

JE - Okay so balik po tayo sa logout, then balik tayo dito, then try naman natin ang kabilang button which is yung log-in. Then, dito naman po is admin@gmail.com and then same muna po since test lang naman tayo. Ayun po siya, ayan.

So makikita po natin na mas marami po siya dito sa left side. Ayan po ang mga abilities ng admin. So first po, actually same lang naman dito yung difference nalang din po dito sa dashboard, earlier is makikita po natin ang recent invoices, so kailan po yung huling date ng invoice, ano yung name ng customer, and yung total price. So hindi po namin yan sinama sa user kasi this is more for the admin lang din naman po talaga. Since sensitive nalang din po yung data na ito and based po sa last na usapan po natin na this is something for the admin side nalang din po, yung pinarequest po ninyo.

Ah so kunyari eto po yung mga accounts, kunyari gumawa nalang po tayo ng account dito.

R - Mga sub accounts na yun or ano?

JE - Opo mga user nalang din po siya.

JE - For example nandiyan po ang kaninang ginamit na account po na yung user@gmail.com, sige punta tayo doon sa create account. Kung gusto natin gumawa ng bago Sige so try po natin gumawa ng create account po tayo, and then pwede po tayo mag enter nalang dito. Sige eto po sir try po ninyo gumawa ng bagong account.

R - Optional ba yan or ano?

JE - Yung middle name po pwede initial nalang po, yung optional nalang po diyan is yung suffix po. You can leave that blank kung wala po.

R - Kahit ano nalang po no?

JE - Opo kahit ano nalang po. And then create account nalang po tayo, and so makikita po natin na success po ang pagcreate ng account, so press lang natin na okay. If we press view accounts makikita po natin na magpapakita po siya. So I believe pwede din natin ito gamitin for the log-in no?



CN - Yeah pero try nalang yun mamaya.

CN - So mamaya nalang po natin yun i-tetest since marami pa pong functionalities.

JE - So before we go to the next one po, keep in mind po yung gagawin na accounts po dito ay only Users lang po, as we believe na dapat isa lang ang admin na account, may isang tao lang ang naghahawak, and you can create as many user accounts as you like.

R - Users lang diba?

JE - Opo

R - Oo, agree naman ako doon.

JE - Tapos i-check mo nga yung edit functionalities ng users account. Okay so edit po natin kunyari eto, so pwede po natin magchange yung mga number ganyan lang naman po. So punta muna tayo sa Create Invoices na module. CJ gusto mo ba ikaw mag explain nito?

CN - Sure kami ni Charles.

JE - Ayun sige.

CN - So yung create invoices po nagbase po kami sa nabigay po ninyong mga excel sheets last time and yeah dun nga po yung pinaka una muna po diyan yung File Name so maglagay po kayo po diyan ng magiging title po, like kahit ano naman po.

AV - Mga pangalan ng invoice usually po.

CN - Tapos after po niyan i-fill up na po ninyo ang mga necessary na details. Tapos yung customer name naman po makikita po ninyo na lahat po diyan po, ay kung ano ang laman ng Customer na.

R - Sila na ang nandoon?

CN - Opo, pero kung may gusto po kayo na kunyari may customer po kayo na hindi mahanap, pwede po kayo gumawa ng bagong customer tapos ilagay po diyan. Automatically naman po, dynamic po yan eh.

JE - Magrefresh nalang din po yan.



- R - Wala bang option dito na blank siya?
- CN - May mga times ba po na may invoice po kayo at wala pong tao?
- R - Hindi I mean, na hindi pa siya, kunyari hindi pa siya bagong...
- JE - Bagong dagdag po?
- R - Bagong customer, so kailangan mong i-add dito sa?
- CN - I-add nalang po ninyo
- JE - I-add nalang po siya, kasi bawal po na walang customer po sa invoice.
- CN - Kasi pinaparequire po.
- R - Hindi, I mean kung panibagong customer wala kang option na i-type muna dito?
- CN - Ay hindi po, dapat po doon, Kasi po magiging masyadong complicated na po
- JE - And purely for invoice lang po siya itong area na ito. So pili nalang po kayo ng kunyari ng kahit sino.
- CN - So after po niyan, diyan po follow lang po ninyo ang instructions may format na po na naka lay-out po diyan na i-follow lang po and additionally po kunyari kasi may mga iba pong invoices na umaabot ng dalawa, tatlo, apat tapos i-click lang po ninyo ang add product po, na para magdagdag po kasi may mga times na bibili ng alcohol at ng hand sanitizer.
- Yung rason po for the product types, same lang po earlier na, hindi po kasi naprovide saamin yung ano, so yung alternative ba-blank lang naman ang iba.
- R - I see.
- JE - Pero usually po, yung mga iba na meron, pagka-pindot po automatically siya so.
- R - Try natin yung ano meron?
- CN - Acid, ay yung Caustic... Caustic Acid.
- AV - Caustic Soda, try niyo po sir.



CN - Caustic Soda yan po ang may mga different product types po.

R - Ah okay yun na yun.

CN - Opo magpapakita naman po siya.

R - Tapos unit price.

JE - Eto po so for the unit price po hindi po kami namigay ng set price kasi po, alam din po namin nagchachange po ang pricing niyo and at the same time may mga times na nagbibigay din po kayo ng discount kasi po nagbubulk buying din po. Set lang po tayo ng unit price, ilan din po ang quantity natin. So makikita po natin automatically na po nagcacalculate po dito.

CN - Pati rin po pwede po kayo mag add ng product po, for the sake lets try po tatlo.

R - Oh okay, ganto pala pag multiple products.

JE - Yes po, and then kung may gusto po kayong alisin you can delete it nalang here.

And then earlier po may na mention po sa discounts, so yun may base price po kayo na eto, you can check po yung checkbox. And then po bigay nalang po kayo ng percent here.

R - Sige so 5%

JE - Tapos yun na po, so base sa sub-total price na 1,000, mababawasan ng discount and the final price po ay nasa pinaka right side.

R - Eto yung preview?

JE - After po mag create.

R - So walang preview invoice? Na...

JE - Ah like before siya mag...

R - Oo before magcreate.

JE - Wala po siya.

R - Sige.



JE - So success naman po siya okay and you can view the invoice nalang din po. Tapos search po ninyo siya sa search function.

CN - Pwede niyo po rin sa invoices. Ayan po.

AV - So download niyo nalang po. Yan po ang makikita ninyo yung invoice. Ganto po ang makikita ninyo sa invoice, so ang format binase nalang po namin siya sa sinend po ninyo dati na mga dito. So ganun lang din naman po ang lalabas kung nag download.

R - Pero pwede naman siya i-PDF diba? Convert siya sa PDF before i-print siya?

JE - Ah, pag nandito na po kayo sa excel.

R - Ah.

JE - Opo, sainyo na po yun kung papaano niyo po siya gusto i-convert. Nakikita naman po namin from the previous meeting na eto na po ang hinihingi ninyong archiving na po, so yun po ang isang way.

CN - Dagdag lang po namin kanina ang functionality po na hindi napakita kanina sa user, which is for example sa InvoiceTest kung may, let's say na hindi na kaya or relevant na pwede na po ninyo i-delete po gamit ng X, kung kanina po napakita natin sa Users account ninyo na pwede merong edit na function, gusto ko lang po makita ninyo na pwede po kayo mag delete. So for example.

R - Delete ko lang yan? Yung invoice

JE - Yes po tapos magrerefresh, tapos wala na po yan sa overall na database po namin yung TestInvoice.

R - Ah wala kayong delete archive?

JE - Ayun po sir na delete na po siya.

R - Hindi I mean wala ba siyang mga...

JE - Ay yung mga stinostore na kung saan...

R - Oo kung saan stinostore ang mga trash.

JE - Ah yun hindi pa po naiimplement.



CN - Palagay nalang po sa recommendation if ever po.

R - Sige po, kasi yun nga kung may magkamaling ma delete, yung mga incidentally.

AV - Opo may mga instances po na...

CN - Kaya nga po ang ginawa namin for admin lang talaga siya, since sila ang may hawak na highest power, kaya ganoon po talaga.

JE - Tapos let's proceed po sa mga na mention kanina, so unahin muna natin sa customer, so yung makikita na po ninyo na difference sa customer kanina sa user and sa customer ngayon sa admin ay sa admin po makikita na po ang email at ang address. We believe na dapat po ganyan kasi syempre sa mga users pwede ka gumawa ng kahit sino sino account at ibibigay mo sa kahit sino sino, syempre diba protection sa consumers.

R - Pati na rin sa company rin.

CN - Consumer protection po and all that, so dapat lang ang mga taong mga nasa highest lang nakakakita nun.

JE - Opo since most especially nalang din po, kayo din po ang cocontact sa kanila, since kayo po ang admin. Provided nalang din po siya here. For the user po more on, makikita lang nila na eto lang pala ang listahan natin. And then again pupwede tayo guma ng...

CN - Maglagay ka muna ang address, for example kasi based po, base lang po sa mga values ng mga invoices na nabigay po ninyo saamin. Kasi may mga invoices po na nakaaddress sa ibang company na walang address. Kaya ganun lang talaga po.

JE - Dito po sa email lagay natin test@gmail.com and sa address lagay lang natin Quezon City. Ayan po pag sinave natin ang changes. Success po, press lang natin okay and then magreflect lang siya agad. Ayan lang naman po siya.

CN - Tapos next po para sa product na view.

JE - Now for yung product naman po natin, ayun po dito nalang din po mapapakita ang product type at yung description, so lets say Acetone po.



CN - Hindi let's say gawa muna tayo ng product, since napakita na natin ang separate parts ng CRUD, Create, I mean yung Read nakikita naman po ninyo as is, so galing sa database nareread.

R - Oo

CN - At din idisplay na po sa mga tables.

JE - So for example po sir gawa na po kayo ng product dito.

R - Um... papaano ito.

JE - Kahit anong product, test lang po siya para ma test po natin siya.

R - Sige lagay ko dito cosmetics...

JE - Tapos kung ilang ang nasa current stock po lagay natin lets say 50.

R - Dito?

JE - Description, kung ano pong description ang gusto po ninyo.

CN - Hindi naman po required yung description.

JE - Skincare, so ayun add po natin. Ayan ang success na po siya, and okay mag refresh nalang din po siya and yun lang din po yun, yung serum. Ayun po nagreflect na po siya.

CN - Tapos maeedit rin po yun, for example mali po tayo na nasulat sa description or sa product type or sa stock po mismo. Pwede po natin siya ma edit diyan.

R - Hm... So dito na nagaano ang mga SKU kanina doon.

JE - Opo yung sa stock and then pwede na po ninyo i-delete. And dito na po ninyo macocontrol ang mga stocks po ninyo. Eto na po mismo yung inventory management side na magpapakita.

CN - Tapos for example, obsolete na yung nalagay ninyong dating product, pwede rin po ninyo i-delete.

JE - I-delete po natin yan na kunyari, remove, yes and delete na natin. Palagi po siyang may "Are you sure" po para sigurado lang talaga. So delete magrefresh nalang po. And then search nalang din po natin just in case na wala na po siya. So pagwala po sasabihin nalang din po na "No matching



records found". So now last is invoice, before po tayo pumunta sa pinaka huli po natin.

CN - So eto na po ang mga invoice natin, may File Name na po siya eto na po yung mga pinaka. Dahil po noong huli ang mga file name po binase lang po namin siya sa pang ilang siya drive.

R - Kung anong sequence na?

JE - Opo, kung anong sequence na po yun. And then kung gusto po ninyo i-sort yan by the file name, or by the dates of the invoices, ayan nalang din po siya.

R - Opo.

JE - Again discount, based nalang din po yan sa kung ano po ang gusto ninyo, and then we can download po and edit the invoice. So kung gusto po ninyo the invoice itself, so kunyari.

CN - Let's move it na sa Sales Forecasting since napakita naman natin ang edit kanina.

JE - Oo since napakita naman natin lahat earlier. So let's move na po dito, eto ang pinaka gusto po namin ipakita sainyo, so yung sales forecasting. Eto yung pinapakita po namin sainyo earlier yung sa or even yung sa start ng year po. Napakita po namin na gusto po namin magkaroon ng sales analytics ninyo, so eto po yung total. So mapapansin po natin earlier doon sa dashboard. Na simple lang po siya na sales parang for this year lang po siya. Pero pagpasok natin dito sa sales forecasting makikita na po natin na oo may 2019 dito all the way to 2024.

So yung rason po na bakit 2019 is para masabi po natin na accurate yung pag calculate niya. 5 years yung agwat po.

R - Ah okay.

JE - So we take nalang po namin, is all the way until 2019. So kunyari po, gusto po ninyo i-isolate natin, pwede nalang po ninyo ipindutin lang natin tapos mawawala nalang din po siya dito. So pwede nalang din po mag interact lang po kayo ngayon dito po sir. Para ma test nalang po ninyo.

R -Ah okay.



JE - Ayun po so makikita po ninyo na nawawala nalang din po yung mga ibang dates. And then pwede parin naman po kayo maghover ang monthly niya na sales. So since sales history lang naman po siya. Na setup po namin na monthly lang po siya makukuha. Pero pagdating na po doon sa forecasting proper, ano na po yun tig-three days na po ang agwat natin.

R - Ang agwat nila

JE - Opo, so kunyari gusto na po natin magforecast, so pipindutin nalang po dito sa baba ang start forecasting.

R - So future na ito?

JE - Opo, so pagnagstart po tayo, ayan nagforecast lang din siya ngayon. Ayan forecasting complete na po siya. So close natin. Ayan na po eto po more on advanced lang po siya, mga mathematical. Yung mga ibig sabihin nalang po nito, is if close po siya sa 0, better for this. Yung rason lang po na bakit mataas po ito is because may bulk buying po sainyo and yung data is usually in millions or hundred thousands. Pag na predict na po na kunyari ng system namin na 100k pero ang actual na data is 500k. May discrepancy parin po siya, pero makikita naman po ninyo sa parang po sa graph may trend line naman po. Tumatama naman po siya. So eto po siya, eto po yung prediction.

CN - Pakita mo ang full-screen functionality ah.

JE - Eto po yung full-screen functionality. Eto po ba yung 2023 lang? Yung prediction na naka.

2023 at 2024 yan, upto 2024 yan yung 2 years.

JE - So eto po yung pagpredict niya po. Nakikita naman po ninyo na gumagana ang full screen and kita naman po ang mga predicted values niya po. Click niyo lang po ESC para ma exit po ang fullscreen.

Eto po yung forecasted sales, eto naman po yung sa future na outside na po siya sa data na pinakain natin. Kaya makikita na po ninyo ang mga dates niya na po almost mga June na po roughly kailan ba ito, mga May po kasi during the time po na wala pa pong data, kasi yung kinuha po namin is until April lang po roughly. Makikita po ninyo ang eto po yung prediction niya, so for the forecasted po yung future, wala talaga po siyang additional line, isa lang po siya.



Kasi yun mga yun...

Kasi future po siya, so parang guide niyo nalang din po yan na okay na baka tumaas lang din po yan. And then...

R - Hindi ba siya pwede mag monthly, dito or every month? 3 days lang?

JE - 3 days lang talaga po, every three days lang talaga siya. And then ayun po, pwede niyo nalang po i-full screen to see. Now escape po tayo, eto po yung parang tabled version ng earlier. So makikita po natin ano yung forecasted na po na sales, ilang yung percent ng growth, so makikita po natin from the last mababawas tayo ng 9% and then 200% for the next one. And then makikita nalang din po natin yun.

So in the next 24 days po yung forecasted natin.

So ayun lang naman po siya so far, and then also po sa Sales History you can also full-screen nalang din po siya, para mas kita lang po siya.

R - Nice nice

JE - And then yun lang naman po siya.

R - So ano lang talaga siya, for inventory purposes lang talaga siya? Yung dashboard natin...

JE - Inventory po and yung dashboard parang pang ano lang talaga siya, overview lang po siya. So rundown lang po. Sa admin sila lang po ang mga nakakacontrol po ng mga accounts, sa invoices, pati sa sales history, sales forecasting. Since si user po hindi naman nila need masyado nun. However po, mamaya sa mga i-papatest natin ipapakita lang din yun sa sales forecasting para makita nalang din po nila. And actually may na prepare lang po kami dito na diba sir earlier hindi nakakita or nagpakita ang sales so ipapakita ko nalang po dito is yung picture po nun. I believe it is this one.

So eto po yung mukha niya. Eto po ang magpapakita po doon. Full screen natin siya. Dito po sa Red makikita natin ang prediction natin and then yung blue is yung actual data.

R - Yung actual natin?

JE - Yes po, so nakita po namin kung tama po kami yung *@!?-_cA po yung bumili nito sa bandang start of the month na ilang million po ata yung



tinaasan nun, so makikita po natin yun sinbai ko po earlier, is kung million po pero yung na predict po ng system is

Mas lower dun sa actual

Malaki talaga po ang discrepancy, pero in terms of the trend line po, kita po natin ang.

R - Same lang naman

JE - Nakukuha po natin siya, opo. As per the recommendation po ninyo. Even though yung system po natin na ito yung sales prediction is more on the admin side, mapapakita nalang din namin siya sa mga ibang user natin, para makita ni user na eto pala ang mga different functionalities ni admin. And then yun nalang din po yung total sa system namin. So ang rason po na tatlo ang laptop namin is doon po sa mga magtetest po madali pagmaraming tao. Madali po pagnagtetest at the same time. And then eto rin po pala yung sa evaluation ng system po.

JE - So dalawa po bale yung survey po namin, mabilis lang naman po siya.

R - So 2 per person?

JE - Opo

R - Okay sige.

JE - Ano naman po siya sir, likert scale. So check check nalang din po kayo dito and then kung may recommendations po kayo. So eto po siya, this one po is based po sa dati pong logbook, so lagay niyo nalang po diyan ang experiences po ninyo, using yung logbook system and i-aabot ko po sainyo afterwards yung.

R - Based on new yan?

JE - Yes po based on new po.

R - Okay lang optional lang naman diba?

JE - Opo. kahit sa department lang po ang dapat i-sulat, yung lang naman po ang mandatory saamin.

R - Eto yung old no?



JE - Opo eto po yung logbook po ninyo. So yung 1 po is yung lowest yung 4 po ang highest. Kahit check check lang po yan.

CN - Ah, ano tawag dito pa comment nalang po kung may mga suggestions na gusto po ninyo makita po.

JE - Or yung experiences niyo po sa logbook na medyo ayaw po pwede niyo nalang po i-comment. So yung rason po na bakit namin tinatanong kung ano ang nasa logbook po ninyo is para alam po namin kung ano po ang experience po ninyo from logbook and then suddenly switching sa system na po talaga.

R - As in sa digital na no?

JE - And then later po sir kung okay lang po sir sa mga certificate of consent po, kayo po ang isa pong admin here other than Ma'am !@#\$, kayo nalang po ang magsisign as witness.

Ayun po thank you, so eto po ang para sa panibagong system po. So ah yung ano po ang amin criteria ay based po namin siya sa ISO, yung world standard nalang din po siya, and also kung napansin po ninyo may signature po siya sa dulo, ano po yun. Ah sign lang po siya na navalidate na po siya namin ng faculty po.

So eto po yung sa panibagong system, same lang po sa department.

R - May grade ba ito sainyo or hindi naman?

JE - Yung alin po?

AV - Para sa system lang naman.

JE - Opo para makita lang din po namin kung ano po ang pwede namin ma improve, opo. Kasi gusto lang po namin malaman po kung ano ang magiging experience ni user na puro ano lang siya kunyari, desulat tapos biglang, full-on uh digital na siya.

R - Ano kaya yung sa RFID?

JE - Actually po yun po ang isa sa nagustuhan namin ma implement sa ano po sir sa RFID, kaso we werent sure po if meron po kayong scanner or if yung



mga product po ninyo may parang mga barcode system or ID system po kayong nagagawa. May ganoon ba po kayo.

R - Ang alam ko wala pa sa ngayon eh.

JE - Yung po ang isa sa mga recommendation po namin sa future research, is mag implement na ng nun kaso...

CN - Kasi po yung balak namin i-scale up from inventory up to logistics na po, to real-time product management.

JE - Kaso ang rason nalang po na bakit hindi po namin nasama is mas malaking investment po siya for both the researcher and the company. Focus nalang din po kami sa mga hinihingi po ninyo which is yung inventory, archiving, and sa sales forecasting nalang din po.

CN - Pati rin po sir if possible po, diyan sa comments/suggestions part ng inventory po pasabi nalang, kasi po nanghihingi po kami ng numerical value for our hypothesis po, parang inventory accuracy, basically i-eyeball niyo lang po. Sabihin ninyo na “ah, kung ginamit ko ito meron akong 98%-99% na accuracy po”, kasi kailangan po namin ma- gauge po yung...

JE - Kasi po let's say ako na po magtatanong sainyo here, since technically interview po ito.

With yung logbook system niyo po ba naensure niyo po ba na tama po palagi ang yung bilang or nakaka experience po kayo na kailangan po ninyo mag count multiple times po talaga within the day para ma-ensure na tama ang bilang.

R - Oo nagkakaroon ng instances kasi hindi naman ano yun, from time to time hindi mo naman na checheck yung inventory pati rin ang logbooks and ang mahirap rin sa logbooks, laging back and forth doon sa historical mo doon sa inventories mo. So unlike dito sa ganitong system. I-eenter mo lang yung, yung inventory na in more of stocks eh, makikita mo na eh, kung meron man. Ah yung pala ang suggestion ko dito is kung ilan ang total na SKU and ilan ang nabawas.

CN - SKU meaning?

JE - SKU is yung ano po sir, yung parang number ng...



R - Number ng...

CN - Stock keeping inventory.

R - So kunyari eto yung month na ano, month of December. Eto yung total na stocks natin for toner, or item. Um for the month of December eto yung nabawas, and eto nalang yung remaining. Na magcacarry over for the next month.

JE - Oh so parang similar lang po siya dito sa sales pero kung ilan yung count po? Basically

R - Opo yung count.

JE - Ahh actually pwede po yun.

JE - And then let's say po, based po doon sa dati niyong logbook system, sa tingin po ninyo ilang percent po ang nakasulat sa logbook yun yung tamang bilang?

R - Tama naman, kasi ano naman eh yun authorized person lang naman ang nakakapagbilang.

JE - Pero mas matrabaho lang po?

R - Mas matrabaho lang talaga.

CN - Kaya ba po ninyo magbigay ng quantity, kunyari mga 90% or "feeling ko sa logbook mga 90%, pero kung ginamit ko ang system na yan baka umabot ng 95-97, 96%"

R - Opo mga ganoong criteria yung range ng ano.

JE - Alin specifically po sir kasi kailangan po namin malaman po. If you said using the system po, I compare po natin sa dati. If hindi po natin i-cocount yung convenience yung actual na accuracy lang po talaga.

R - Kung actual accuracy, um siguro magrange ka sa 80-90 or 95% sa logbook system, kasi lahat lahat naman accurate ang ineenter namin bawal naman siya magkamali. Pero within this one pwede ka mag range from 90-99% kasi ikaw lang naman yung nakakaaccess diba?

JE - Opo si admin lang talaga.

R - Tsaka safe and secured naman siya.



CN - Ah tawag dito, hindi pala namin na mention po since na bring up na po natin ang security. Dito po sa accounts po, syempre tawag dito. Sa practice namin sa CS at sa IT field. Kung magkekeep ka ng password syempre bawal mo yan i-store as yung string or yung the same alphanumerical values sa database. Instead before po namin i-pasok yan sa database dumadaan po yan sa encryption, na parang.

JE - Parang ano po siya, parang yung password mo is test123, pag pumasok na po yun sa database namin, hindi na po siya test123, naka ano na po siya naka specific na encryption po siya.

R - Ah okay okay

CN - So at least on that side safe din po siya from mga data breaches.

R - Okay okay.

JE - Ayun po and then as you said po 90-99% ang accuracy confirming, and then doon po sa sales forecasting may comment ba po kayo here?

R - Sa sales forecasting, ayun lang naman kasi ang sales forecasting lang naman ay hindi natin masasabing accurate siya, so nagbabase lang naman siya sa mga previous na data. So yun lang naman ang concern ko diyan, although hindi naman siya concern, mga observation lang naman.

JE - Pero para sainyo po sir, with the system na ito, specifically yung forecasting.

R - Oo, sales forecasting ano yan sobrang helpful sa company kasi makikita mo na agad, kung alin yung month yung ano mo, kung um, malaki ang sales mo kung mababa ang sales mo and naman. So ayun lang naman. Maganda siya, um maganda siya for... for planning na maimprove, mo yung pagmamarket mo, sa in the future diba. Kunyari from the last month, ganito yung, eto yung ano mo, but naforecast na ng system kung mababa paba yung this month and the next month, and then makikita mo na talaga, it will help you na to develop yung pagmamarket mo ng mga sales natin.

JE - Okay po sir yun lang po so far ang system namin and thank you po for the interview and also bale po end ko po muna yung recording



TRANSCRIPTION OF THE SELECT PARTS OF THE DEMONSTRATION AND SURVEY ON THE OTHER ADMINS

JE - Julian Kristoff Encarnacion

CN - Cedric John Nedtran

AV - Alvin Charles Villanueva

V - Sales Representative Admin

R - Quality Assurance Admin

T - Production Admin

G - Marketing and Production Admin

C - Production Supervisor

Evaluation and Accuracy Score of the Sales Representative Admin

T - Siguro maganda kung ang nakaka edit lang si admin lang diba? Yung pinaka manager, admin head, no?

JE - Ginawa po namin here is one admin lang.

...

JE - Lets say 5 na po ang nasa Acetone and i-sasave na natin ito, as you can see mag rerefresh lang po and then mag rereflect na po siya. And then same nalang din po earlier yung pinakita sa add invoice meron rin pong add product. Ganoon nalang din po siya.

T - Tapos nagaaauto nalang siya nababawasan kapag ginamit doon sa paggawa ng invoice.

JE - Ah hindi pa po siya, kasi kailangan pa po i-double ang pagadd, pero pwede po namin i-add po siya sa recommendation

T - Ay oo para mas maganda siya. Kasi kung gumawa ako ng invoice, halimbawa kay !@#\$%^ tapos umorder sila ng Air Fresheners and then



meron akong 200 na stocks, kumuha sila ng 100, yung nagleless than na po. Para nakikita lang po.

JE - Ay sige po add lang po namin yan sa recommendation

T - Maganda po ang ganoon.

JE - Kasi po for now ang nagawa lang namin ay seperate lang si invoice on its own, para hindi po siya magka issue, ng what if mas mababa pala ang current stock. Pero i-add nalang po namin yun sa recommendation. And then let's go na po sa sales forecasting

...

JE - So kunyari dito po ano po ang mga experiences po ninyo sa logbook, mas madali ba makita, mas madali ba...

T - Hindi, kasi kung naka data siya CTRL + F mo lang tapos nandoon na, hindi na mauulit yung anong year? *laughs*

JE - So lagay nalang po ninyo sa experiences po ninyo, yung 3 po is satisfactory and yung 2 po is below average and 1 po is yung needs improvement. So kunyari kung ayaw na po ninyo i-1 nalang po ninyo talaga.

So last question nalang din po pati na rin kung may comment po kayo doon sa traditional system po.

T - Yung traditional system?

JE - Opo kung may comment po kayo sa logbook po or parang isang thing na ayaw po ninyo sa logbook po.

CN - Gusto po namin malaman po yung...

JE - Kung ano po ang throwback po noon.

V - Sulat ng sulat

R - Lab na lab ni ma'am yun *laughs*

V - Sulat ng sulat, iba talaga pagnageenter nalang

JE - Pwede po ninyo ilagay po yun sa comments po. Eto po yung nasa panibagong system na po, yung pinakita po mamaya.



CN - Tara swap na tayo. Ako muna po ang sainyo po. Siya kasi po ang mas magaling po magexplain po saakin diyan.

T - Ikaw si?

CN - Nedtran po, balak nga po namin kung na click po yan at the very least makikita po ninyo kung sino sino ang nagcode, sa una po balak namin ang facebook, pero much better po ang linked in muna, para mas professional.

T - Ano ulit ang name mo?

CN - Nedtran po.

...

CN - Dito po nakikita po ninyo na hanggang tingin lang po talaga, walang add, walang delete, at walang edit.

T - Pero meron akong extra features, kasi maganda ito ah. Ako halimbawa meron akong invoice, pero hindi ko alam kanino ipeprepare kaya gusto ko kapag cinlick to lalabas ang mga products na laging binibili.

CN - Ah may pop-up?

T - So para magkaroon siya na idea, na ah, eto yung laging kinukuha niyang room spray.

CN - Ah okay so sige sige , palagay nalang po sa recommendation na may different view po no, or may mga different profiles po no, based on the products typically bought.

T - Oo additional lang naman, kasi diba kung may mga bago, tapos binigay namin ng order na ah kunin mo ang laging binibili nila, minsan nakakalimutan nila kung ano ang brand yun. So pagcinlick niya. Ah eto siya lang ang kumukuha ng...

CN - So parang sales history po no per customer?

...

T - So ano yung safeguard namin na hindi siya, ma prevent or yung accidental or yung unauthorized modification.



CN - Yung way po niyan for example po miss, is titingin ka at gusto mo mag-edit pero oh no na click ko ang X. Pero buti nga lang may confirmation pa pala. Same rin po dito sa forecasting pagnagsimula na po bawal na po i-stop, kaya pinapacconfirm po. Kaya po sinasabi na Are you sure about kung gusto mo i-tuloy ito?

...

T - Ang maganda rin, aside from admin parang meron kang interface, na naglologin din si owner. Meron ganoon ba dito sa ano natin?

CN - Pwede po namin ilagay po yun, like for example may pangatlong pong type of account. Na si owner siya na talaga ang pinaka master.

T - Nakikita niya ang data na lumalabas

CN - Opo nakikita na niya ang mga passwords at ang mga data.

T - Nakikita niya na rin kung si admin nag delete, yung history

CN - Opo para ma keep track niya po, para makita niya po yung galaw ng company.

T - Kung ano ang mga na-aadd, na leless.

CN - Sige po palagay nalang po yun sa recommendation po.

...

CN - Encar yung inventory accuracy

JE - Yung luma or yung bago?

CN - Both.

JE - Okay i-comment niyo nalang po para sainyo po sa tingin ninyo ilang percent po accurate ang logbook, compared sa actual na count sa warehouse.

V - Hindi accurate kasi nga mahirap

JE - I-comment niyo nalang po yun Ayan sige po and dito po ang nasa inventory system po namin.

...



CN - May specific part po kami sa papel that deals with the hypothesis, meaning mga hula hula lang po. So kayo po sa hula hula po ninyo,

JE - Estimation

CN - Sige estimation, mas formal na term. Sige po sa traditional i-eyeball po natin sa mga traditional na logbook “feeling ko mga 91%, pero feeling ko pagnaimplement na ito sa company operations baka umabot pa ng 96%”. Pasulat nalang po diyan sa comments.

T - Para saakin yung system syempre mga 100% yan

CN - Thank you po, tapos dito po ang sa logbook po. Tapos kung may gusto po kayo i-airout na greivances po sa logbook palagay nalang po dito.

T - Dito sa logbook feeling ko mga 50%, kasi minsan mahirap pa maghanap sa mga files.

CN - Opo kasi isipin mo rin ang mga papel, kung hindi maganda ang taguan, medyo magiging mabasa basa.

T - Kasi halimbawa nagkaroon ng sunog at ng baha, wala na, wala na lahat na yun. Pero kung may system. Kahit na sa ano ka. Pero etong system, wala ba siyang kinakailangan na maglog-in, like sa office lang? Or pwede kang...

CN - Kahit saan po, since web application po ito, and since ako po ang front end developer ng front developer po nito. May mga mobile na view po at pupwede po ito ma view sa cellphone, sa laptops, at sa PCs.

Pwede niyo rin po dito i-add po, kasi may mga accounts accounts po diba? Pwede po ninyo i-add dito yung naisip po namin sa una, which is yung communications. May mga accounts accounts diba? Pwede po kami mag add dito ng notifications, messaging, at ng ganoon po. Parang naging social media po, pero I think that’s para sa next set of researchers na po na gagawa. Kung gusto po ninyo i-recommend yun.

T - Mas maganda yun since halimbawa may naglogin sa ano, may magnotify saakin na may naglogin po. Meron bang ganoon?

CN - Pwede po naming gawin yung...



T - Kasi halimbawa nagpasok siya ng entry, makikita ko naba agad yun sa notification na meron na nagpasok ng item.

CN - I mean doable naman po yung since meron po kaming ganoon po kung saan po kami gumagawa ng schoolworks online. Meron po parang activity page, na kungyari, this user did this, this user looked at that. I believe kaya po namin i-set an eventlistener kung may... Basically lang po sa owner na may nadelete po lahat ng admins, pero yun lang. May bagong customer pwede natin i-set na makikita na ng lahat, pati na rin ang mga users yun. Yeah doable naman po yun, pero pasulat nalang din po sa recommendation, para ma take note po namin. Yung messaging, activity logs, and notifications.

...

JE - Actually originally po ang system po namin, a lot more on the logistics side po, pero ang recommendation po saamin ng panel, considering yung time namin and yung system, magfocus lang po kami sa inventory aspect, and yung forecasting po and yung additionally po yung gusto po ninyo which is yung archiving yung nalang din po ang focus namin. So yun lang talaga po ang laman ng system po namin.

V - So for future nalang kung talagang gagawa po kayo ng actual na, balikan niyo lang kami, kasi kailangan na kailangan na namin yun. At tsaka pag kayo ang gumawa ma mura mura. Syempre kasi pag dinala namin ito sa iba mga kalugdan na.



Evaluation and Accuracy Score of the Marketing and Production Admin

G - Ano ulit ang isusulat ko? Nakalimutan ko kasi.

JE - Yung how many percent accurately, do you think it will be, like a 95% accuracy, 98.

G - Dapat 100 no?

V - Oo 100, parehas tayo ng sagot *laughs*.

JE - Pwede naman po 100%

G - Kasi you have to be ano, I mean kasi sa tao naman ang gagawa nito eh.

V - Minimal lang naman kasi sa system.

G - Kasi sa tao lang naman ang pagcalculate eh, 100 as long as well maintained.

JE - Okay thank you so much for answering po.

Inquiring on what information to document and what to omit.

JE - Hindi naman po namin siya ipapakita, ano nalang po kapag i-pepresent po, kunyari ang mga names po ng mga product, hindi po namin siya ipapakita.

V - ABC ABC lang po ha.

CN - Opo mga XYZ.

V - Uh huh mga XYZ lang.

AV - May switch naman po kami na gamitin ang dummy data.

V - Okay mga dummy data, ganoon nalang.

CN - So tawag dito, may suggestion po saamin si adviser po kung kaya, or pupwede ba pong isang invoice lang ang papakita, yung i-crecreate namin and edit, delete. Okay lang ba po kahit yung isa lang po. Galing sa actual na sales or bawal talaga?

V - Bawal talaga.



- T - Bawal kasi yun talaga ang sales invoice natin.
- AV - Ano nalang yung papalitan, pero yung layout same lang naman.
- JE - Yung XYZ?
- V - i-XYZ niyo nalang.
- JE - Company X nalang tayo.
- R - Pwede niyo naman i-edit yun.
- CN - Gagawa nalang talaga po ng dummy invoice, sige.
- JN - Sige po kasi yun po ang rason bakit po binigay po ninyo saamin, kasi confidential na po talaga.
- R - Kasi kung ganoon, gagawa tayo ng NDA, na Non-Disclosure Agreement
- V - Pero as kay Ma'am !@#\$\$%, no need na mag NDA basta wag lang i-present yung mga name, mga company name and yung mga customer.
- JE - Sige po, i-withdraw lang po namin yan sa paper.
- CN - Sige po since nandito na po, kasi parang eto na po ang dati ko pong iniisip, kung sa pagsulat ng papel. Kung pupwede po namin kayo i-name drop, or gamitin po namin ang Chemical Manufacturing Company in Quezon City.
- V - Pwede naman po yun.
- CN - Sa mismong title?
- V - Pero nasabi na yun ni sir !@#\$\$% diba?
- JE - Pero in terms po of the presentation, if tinanong po kami yung name ng company...
- V - I-confirm mo nalang
- AV - Pwede naman ata kasi nakalagay na ang name sa paper.
- V - I-confirm mo nalang just to be sure.
- JE - Pero for sure po ang invoices, yung sensitive information...



CN - Bawal na talaga, bawal rin tayo gumamit ng kahit isa.

V - Ganito kasi yun, yun mga parang secured yung data, kung secure ba yung system or whatever, yung mga third party

CN - yan po ang ginamit naming Bcrypt hashing para encrypted po ang mga passwords kung gagawa ng user ng account.

POST SURVEY DISCUSSION

JE - Julian Kristoff Encarnacion

CN - Cedric John Nedtran

AV - Alvin Charles Villanueva

V - Sales Representative Admin

R - Quality Assurance Admin

T - Production Admin

G - Marketing and Production Admin

C - Production Supervisor

JE - Sige po ma'am parang pang-transcribe lang naman

T - Pero malaking tulong yung data na yun sa owner kasi halimbawa, nagkaroon kami ng month end tapos meron kaming projection na sales, so halimbawa sa side naman namin sa production. Magkakaroon kami ng projection na eto yung, halimbawa yung mga products na ganito yung pwede naming na project.

JE - Opo

T - So doon sa forecasting na yun versus doon sa actual, kaya ang sinusuggest namin from doon sa projection niyo may actual na nakikita din. So big help yun sa owner.

CN - For Clarification po, kasi for now ang forecasting namin is kabuuang sales, so gusto po ninyo ang individual products may sariling forecasting?



T - Hindi, pwede rin naman yung sa sales kasi malaking help rin naman, halimbawa si admin, si sales meron silang forecasting ng sales nila kung na target ba nila yung sales sa forecast na yun or hindi kung saan bumabang month, saan tumaas...

JE - Kung tumama siya?

T - Oo, kung tumama siya.

JE - Siguro po disclaimer lang din, hindi po namin na pepredict din yung mga kunyaring events na pagmay outbreak ng kung ano po, yung mga outside na po more on data lang saamin. Based on patterns lang kasi.

T - Pero naman kasi diba, once na nainput namin like for example nag forecast kami ng ganito 3 M, for the month of ganito, ang sales namin ganito. So versus doon sa actual namin na kung gagamitin namin siya na mga invoice, pasok kami ng invoice, stocks, ilan ang lumabas saamin, nakikita na rin yun mag aactual na rin siya.

V - Oo in a way kasi may data rin siya.

T - Kaya malaking tulong rin yun na merong ganoon. Kasi sa owners side makikita nila kung nagfunction pa ang admin or ang sales sa production kung sakali addition na and ganoon.

JE - Yung lang din po kasi ang gusto po naming mapakita sainyo. Kung okay naman ang paggamit

T - Pero maganda po yung ganun niyo

CN - Ang naiisip ko nga po baka kulang pa sa views, kasi yung nag-email po, may messaging pa, may widgets, may mga shortcuts, may mga marami pa kaming balak po

JE - He wanted to-

CN - Cinut nalang talaga po due to time constraints.

JE - And ano nalang po kasi. Uh we want to focus nalang din po talaga to focus lang din po sa pinaka goal namin which is yung inventory...

V - Yung inventory and yung invoice?



JE - Yung invoices, yung archiving po, and yung sales na analysis yung lang po yung gusto namin.

CN - Opo, yung parang pang dagdag lang naman yung user accounts, let's say like yeah kung gagamitin naman ng actual na company, bawal lang naman nagiisang account, dapat may different levels of authority.

V - Access?

CN - Yes

JE - So yun lang din naman po ang isa na namention po ninyo, that we might add sa future, um after the defense na po yun. Would be yung may owner level, so si owner may oversee niya lahat ng mangyayari.

CN - Opo siya yung pinaka mataas na nakikita niya lahat ng activity logs, like si Admin gumawa nito, ah si User may, ay may na tawag dito, may bagong user account na cretead, ah may dinelete.

V - May notification na doon sa?

JE - Yes, parang overview po siya.

CN - Ang notification mabilis ma flood yun, naiisip ko po parang alam mo sa schoolbook...

JE - Yung logs?

CN - Oo, yung logs basically parang table na laging pinapa apend-apend-apend na nandoon na lahat tapos may isang click na view all. Basically parang notifications lang na hindi sa pinaka top-right, parang nasa isang portion ng dashboard.

R - In addition, siguro doon sa pinaka master access, meron siyang option na oh eto si admin na ito, eto yung gagawin mo, mag edit, mag change, and then bawal ka mag delete, parang drop down lang na ano na category ang pwedeng gawin

JE - Ah, okay.

R - So eto si user view lang, view purposes only. Si admin 2 na ito lang ang gagawin niya pwede siya mag invoice and pwede siya mag...



JE - Ah, so may specific po na user, na kunyari ako po, pwede lang ako magview nito

R - Opo

JE - Ah, okay so.

CN - So mga checkbox, checkbox lang po diba? Kunyari set this account, mga parang mga account restrictions, ay parang mga...

JE - Parang may slider ka, na pipili ka ng kung ano mga pwede gawin ng mga...

CN - Like mga account restrictions, ay oo yeah kasi syempre magiincorporate na tayo ng mga different departments, syempre may mga um, Accountancy Admin, Product Admin, and syempre ganoon. Iba-iba ang levels of authority nila. Iba-iba ang mga use cases.

JE - Pero ayun po, like I mentioned nalang din po, kasi for the sake ng thesis pati ang nasa oras. Focus lang din po namin ang pinaka-primary na gusto which is yung sa inventory, sales, and pati yung invoice, yung archiving.

V - Pero maganda yung ano nila, kasi kung ganyan ngayon palang, mas gaganda kung lalo na madedevelop.

CN - Ang maganda Encar, kung after graduation habang nasa nagtatrabaho ka sa BGC gawin mong side-hustle.

JE - So again maraming salamat po for the time and sa participation po.

CN - Message nalang siguro po sa Viber for the updates po.